CHAPTER 6

DETERMINATION OF MODEL CONSTANTS

INTRODUCTION

In the two general models, values for a number of constants defining the influent wastewater COD and TKN characteristics and the biological kinetics and stoichiometry need to be known. In the programs, **UCTOLD** and **IAWPRC**, default values have been assigned to these constants. For the constants defining the influent wastewater COD and TKN characteristics, the default values are averages determined for South African municipal wastewaters; for the kinetic and stoichiometric constants, the default values have been determined by direct measurement and by simulation of the responses measured for a wide range of experimental activated sludge systems.

In this chapter guidance is given for the determination of the influent wastewater COD and TKN characteristics, and on the kinetic and stoichiometric constants which appear to change with different wastewaters and/or systems.

INFLUENT COD CHARACTERISTICS

Knowledge of the influent COD characteristics is of importance in determining carbonaceous oxygen demand, sludge production, denitrification and effluent COD.

The influent COD characteristics can differ greatly between wastewaters. It was shown in **Chapter 2** that the COD of wastewaters can be divided into two main fractions, unbiodegradable and biodegradable (Fig 6.1).

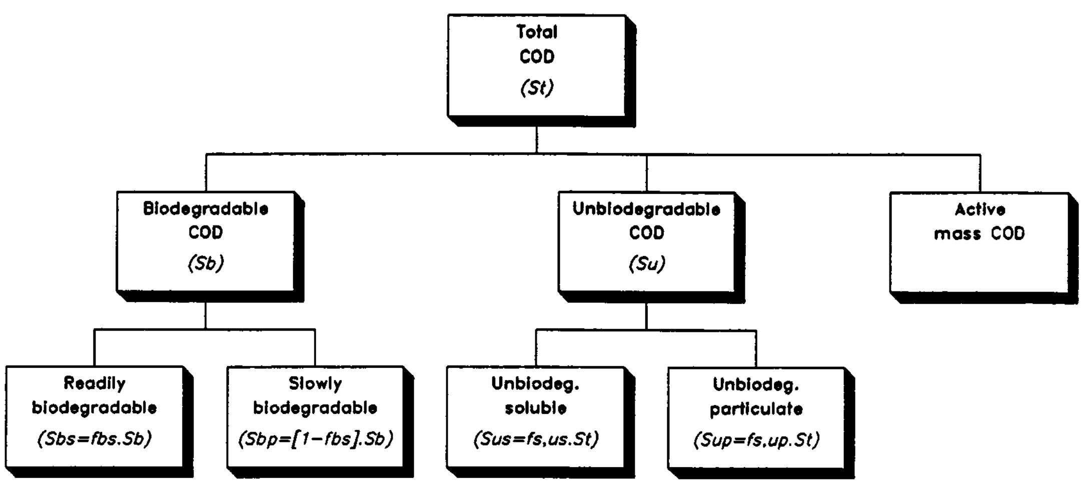

Fig 6.1: Division of influent COD into its constituent fractions

The unbiodegradable COD has two sub-fractions; (1) unbiodegradable particulate (UPCOD) and (2) unbiodegradable soluble (USCOD). The biodegradable COD also has two sub-fractions; (3) slowly biodegradable (SBCOD) and (4) readily biodegradable (RBCOD), this latter subdivision is made on the basis of *biological response*, not on physical separation. Procedures are presented for determining RBCOD, USCOD, UPCOD; subtraction of these three COD fractions from the total COD gives the SBCOD. Following the IAWPRC recommendations, an additional influent COD fraction is included in the models, that of influent biological (active) mass, see Fig 6.1. However, this COD fraction can be determined only by trial and error fitting of simulated to experimental data (**Chapter** 2) and accordingly procedures for its determination are not included in this Chapter.

Determination of readily biodegradable COD (RBCOD)

The method recommended and presented below for determining RBCOD is the flow through activated sludge system method. (Batch test methods also are available for determining RBCOD, see later).

Basis of test

In experimental investigations into the dynamic response of activated sludge systems (Ekama and Marais, 1979; Dold *et al.*, 1980), it was found that in aerobic completely mixed single reactor systems at short sludge ages (1,5–3,0d) under daily cyclic square wave loading conditions (feed 12h on, 12h off) there was a precipitous drop in the oxygen utilization rate on termination of the feed period (Fig 6.2). This behaviour was hypothesized to be due to the RBCOD in the influent – during the feed period the RBCOD is utilized at a very high rate and virtually completely; at feed termination addition of RBCOD also ceases causing a concomitant precipitous drop in the OUR. Measurement of the drop in OUR at feed termination therefore provides a means for estimating the mass concentration of RBCOD in the influent.

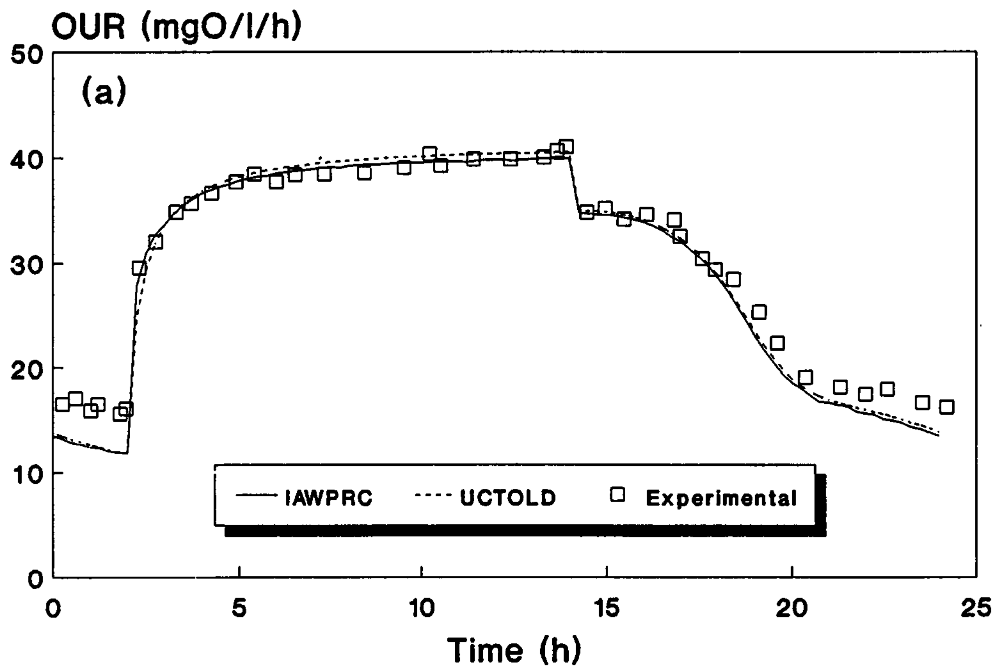

Fig 6.2: OUR-time profile in single reactor completely aerobic system at 2,5d sludge age under daily cyclic square wave loading (feed 12h on, 12h off); note precipitous drop in OUR on feed termination.

Experience with the test, and with theoretical modelling exercises, is that in order to obtain a clear definition of the precipitous drop in OUR, a short sludge age and a daily cyclic square wave loading pattern (feed 12h on, 12h off) must be imposed. A sludge age of 1,5 to 3d usually is satisfactory; for the purpose of the test knowledge of the exact sludge age is not essential, the only requirement being that a clear definition of the precipitous OUR drop is exhibited experimentally.

Experimental set up
A single aerobic reactor completely mixed independently of aeration, a settler, feed and underflow recycle pump channels, and an oxygen meter connected to a strip chart recorder, are required (see Fig 6.3).[1]

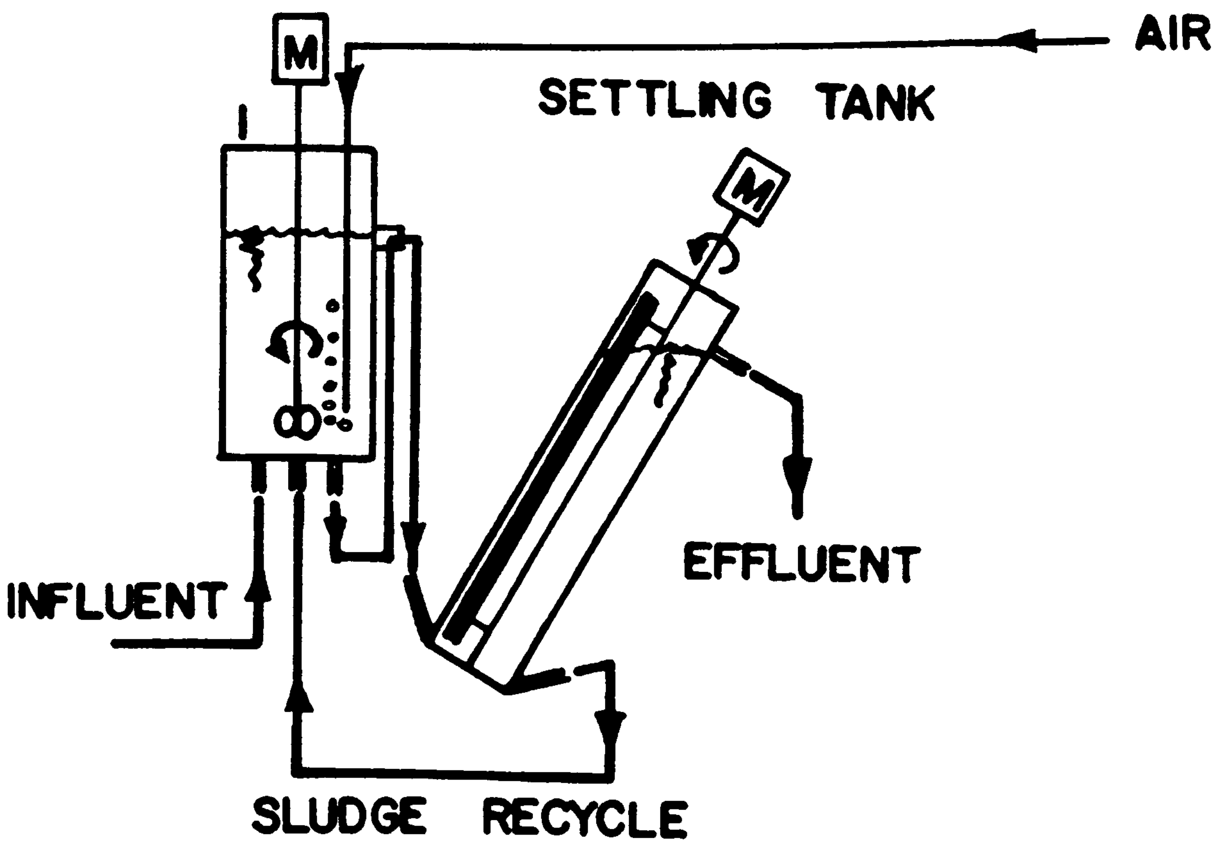

Fig 6.3: Sideview of experimental set-up for RBCOD determination.

[1]Details of reactor, settler and pumps can be obtained on request from Water Research Group, Dept. Civil Engineering, University of Cape Town, Rondebosch 7700, South Africa.

In the set up and operation of the system, the following points warrant notice:

Inlets and outlets from aeration reactor: All streams to and from the reactor should enter the reactor below the liquid surface, preferably at the reactor bottom as shown in Fig 6.3; surface outlets are to be avoided as these may block and/or filter out to a degree particulate material in the effluent flow.

Feed: A batch of the wastewater to be tested sufficient for 1–2 weeks feed to the system should be collected; this is to overcome acclimatization problems and to ensure steady state is maintained. The wastewater batch must be stored at 4°C in stainless steel or plastic tanks to minimize RBCOD utilization during storage. Daily, the wastewater in the storage vessel is thoroughly mixed. It has been found that mixing can be best achieved by using a flat round disk attached at right angles to a long rod; by picking up and pushing down the rod mixing is readily achieved without significant dissolved oxygen (DO) entrainment. A volume of the mixed wastewater is drawn off and diluted with tap water to the desired feed COD concentration; this is placed in a feed reservoir also at 4°C. The contents of the feed reservoir should be mechanically mixed slowly and gently in order to keep particulate matter in suspension, but to limit DO entrainment. Covering the reservoir contents by a floating disk effectively restricts DO entrainment. Daily the reservoir and feed pipes must be cleaned in boiling water. Also, once a week the cleaned feed pipes are dumped in a hypochlorite solution. These procedures are necessary to prevent growth of *Sphaerotilus natans* on the walls of the influent feed line. These organisms readily proliferate in this substrate rich low DO environment. If this proliferation is not controlled, the mixed liquor in the reactors is continuously seeded with *S.natans* which may give rise to bulking (Gabb *et al.*, 1989).

Temperature: The system should be operated at a constant temperature, preferably in a temperature controlled room, at say 20°C. With the cyclic feeding pattern temperature changes may take place in the aerobic reactor due to the low temperature of the feed; these can be overcome by passing the feed through a coil in a warm water bath prior to discharge to the aerobic reactor.

Sludge age control: It is strongly recommended that the sludge age is established by hydraulic control. In this procedure, to maintain the sludge age at the desired value the appropriate mass of mixed liquor must be wasted every day. If the reactor liquid volume is say 10ℓ and the desired sludge age 2,5 d then 10/2,5 = 4ℓ of mixed liquor must be wasted from the reactor daily and the volume wasted replaced with effluent. With such a large volume fraction to be wasted, wasting should be done at least 2 times/day, in this case 2ℓ in the morning and 2ℓ in the afternoon, say. Sludge wastage during the feed period should always take place at least about 4 hours before a RBCOD test is done in order that the OUR has time to regain steady state. Prior to each sludge wastage the stirrer blades and the walls of the reactor must be thoroughly brushed to dislodge adhering biological growth. (About every 2 weeks the mixed liquor must be emptied and stored under aeration while the reactor, settling tank and connecting tubing are thoroughly scrubbed in boiling water). Under no circumstances should the MLVSS (or MLSS) concentration *per se* be controlled by say, adjusting sludge wastage; this creates considerable uncertainty in the system sludge age.

Suggested loading rate: The COD load on the reactor should be such that the OUR (excluding nitrification which may or may not occur at the short sludge age depending on the maximum specific growth rate of the autotrophs, $\hat{\mu}_A$) is greater than 25 mgO/ℓ/h. This can be achieved if the feed rate, COD feed concentration and reactor volume are such that the load during the feed period is greater than 2 400 mgCOD/ℓ reactor liquid volume/day. Thus, if the feed concentration is about 350 mgCOD/ℓ and reactor liquid volume is 10ℓ, the feed flow rate Q prior to feed termination should be greater than $2400 \cdot 10/350 = 68{,}5$ ℓ/d.

Reactor oxygen concentration: The dissolved oxygen (DO) concentration in the reactor must be maintained at a level that does not inhibit the biological processes, $1 < DO < 3$ mgO/ℓ. Fine bubble aeration is preferable, to improve oxygen dissolution. Aeration and mixing must be independent so that during air off periods the mixed liquor is kept in suspension.

Nitrification: The degree of nitrification which takes place in the unit does not affect the RBCOD determination. The reason for this is that at the short sludge age at which the determination is done, the oxygen utilization for *nitrification* is the same before and after feed termination, due to the high ammonia concentration in the reactor and effluent. If the ammonia concentration in the reactor and effluent before and after feed termination is low (< 5mgN/ℓ), the change in the oxygen utilization rate for nitrification may lead to errors in the RBCOD determination. In this case it is necessary to reduce the sludge age or supplement the feed with ammonia.

System start up

From start up, operate the unit under constant flow and load conditions for at least 3 sludge ages to allow the unit to reach steady state - steady state is achieved when the day to day oxygen consumption rate and reactor MLVSS concentration show approximately steady values. Although not essential to obtain the correct RBCOD fraction, it is advisable to do COD and N mass balances on the steady state data (method given below) to give an indication of the accuracy and reliability of the experimental measurements. When steady state is achieved switch the unit over to daily cyclic square wave loading conditions by discharging the daily volume of feed over say 12 hours instead of 24. (Keep the feed volume and COD concentration constant; simply increase the flow rate). Operate the unit under square wave conditions for about 2 sludge ages. Note that now the oxygen consumption rate will vary over the day but the MLVSS concentration should not change significantly.

Should the influent composition change substantially, or if, for example, glucose is added to influent, the system must be run for a further 2 sludge ages before RBCOD measurements commence. Experience has shown that with change of substrate, even if readily biodegradable, an appreciable time may elapse before organisms adapt to the new substrate.

RBCOD measurement

The RBCOD concentration is determined from the oxygen utilization rate (OUR) drop at feed termination. This OUR drop is measured as follows:

(1) At some time at least 6 hours after commencing the feed, measure the OUR by raising the DO to about 6 mgO/ℓ, switching off the air supply only, and monitoring the rate of decrease of DO on a recorder until the DO reaches 1 mgO/ℓ, whereupon the air is switched on again. Repeat measurement of the OUR every 20 to 30 minutes for about 2 hours before the feed is terminated.

(2) Switch off the *feed and underflow* recycle to the reactor and again raise the DO to about 6 mgO/ℓ. About 1 to 2 minutes after the feed has been stopped again measure the OUR, and every 20 to 30 minutes for about 2 hours thereafter. The underflow recycle must be switched off otherwise the sludge in the settling tank is added to the reactor disturbing the 'steady state' VSS

prior to feed termination, which leads to slightly too high OUR's after feed termination.

(3) Plot OUR for the periods of about 2 hours before and after the feed was terminated. Estimate the average OUR before and after the feed termination.

Instead of manually determining the OUR, as described above, an electronic automatic system has been developed at the University of Cape Town (Randall *et al.*, 1991). This system performs all the manual functions, i.e. switches the air on and off between selected lower and upper DO concentration limits; during the air off period it stores the DO-time values, discards those values close to the upper DO limit and via linear regression fits the best straight line to the remaining DO-time values, the slope of which defines the OUR. The OUR's determined in this manner are stored and can be reproduced off-line on a computer.

From the measurements, the following information is used to calculate the concentration of RBCOD in the influent feed:

Q = feed flow rate (ℓ/d) during the feed-on period
V_p = reactor volume (ℓ)
OUR_b = average OUR before feed termination (mgO/ℓ/h)
OUR_a = average OUR after feed termination (mgO/ℓ/h)
S_{ti} = total influent COD concentration of feed (mgCOD/ℓ)
$\Delta OUR = (OUR_b - OUR_a)$ (mgO/ℓ/h)

The influent RBCOD concentration (S_{bsi}) is given by

$$S_{bsi} = (\Delta OUR \cdot V_p \cdot 24)/(Q \cdot 0{,}334) \qquad (\text{mgCOD}/\ell) \tag{6.1}$$

where
24 = number of hours per day
0,334 = conversion factor for COD to oxygen. This factor implies that for every unit of COD utilized by the micro-organisms, 0,334 units of COD are passed to the electron acceptor oxygen and therefore reflects oxygen consumption; the remaining 0,666 units of COD are converted to new organism mass (see WRC, 1984).

As input to the computer programs the fractional constant f_{bs} is required which is the fraction of influent RBCOD (S_{bsi}) with respect to the influent biodegradable COD (S_{bi}), and is given by

$$f_{bs} = S_{bsi}/S_{bi}$$

$$= S_{bsi}/[S_{ti}\,(1 - f_{S,us} - f_{S,up})] \tag{6.2}$$

where
$f_{S,us}$ = fraction of total COD which is unbiodegradable soluble
$f_{S,up}$ = fraction of total COD which is unbiodegradable particulate

Procedures for determining $f_{S,us}$ and $f_{S,up}$ are given below.

Example calculation
The following data were obtained from an experimental unit operated on primary settling tank effluent at the Goudkoppies plant, South Africa,

Q $= 50\ \ell/d$ for period prior to feed termination
V_p $= 7{,}5\ \ell$
OUR_b $= 24{,}1$ mgO/ℓ/h
OUR_a $= 20{,}1$ mgO/ℓ/h
S_{ti} $= 370$ mgCOD/ℓ
ΔOUR $= 24{,}1 - 20{,}1 = 4{,}0$ mgO/ℓ/h

From Eq (6.1),

$$S_{bsi} = (4{,}0 \cdot 7{,}5 \cdot 24)/(50 \cdot 0{,}334) = 43{,}1 \text{ mgCOD}/\ell$$

Now, $f_{S,us}$ and $f_{S,up}$ were determined independently (for method see below) to give values of 0,09 and 0,04 respectively. Accordingly, from Eq (6.2)

$$f_{bs} = 43{,}1/[370(1 - 0{,}09 - 0{,}04)] = 0{,}13 \text{ mgCOD/mgCOD.}$$

Determination of unbiodegradable soluble and particulate COD (USCOD and UPCOD)

With regard to the unbiodegradable COD fractions, even though these are presumed to be unaffected by biological action, determination of USCOD and UPCOD is important for accurate description of the activated sludge system; the former passes out with the effluent as soluble COD, whereas the latter is enmeshed in the sludge mass, accumulates in the system and is removed via the daily waste sludge. Accordingly, the USCOD fraction sets the lowest filtered effluent COD concentration the system can attain (provided unbiodegradable soluble COD is not generated in the system to any significant degree, see later) and the UPCOD fraction affects the sludge production. Measurement of USCOD is direct; UPCOD can be estimated only via the hypothesized model and in this regard it has meaning only in terms of the model structure.

Experimental set up
Two laboratory-scale activated sludge systems are set up to operate at two different sludge ages longer than 5 days - say 10 and 20 days. Each system consists of a single fully aerobic reactor completely mixed by an independent mechanical mixer, a settler and feed and underflow recycle pumps (as for the RBCOD measurement unit). In the set up and operation of the systems the procedures set out above for determination of RBCOD (i.e. brushing of reactors, daily cleaning of feed tubes, etc.) should be followed with the following alterations and additions:

Feed: Collection and storage of the feed is as for the RBCOD system described above. Each of the systems must be fed a constant mass of COD each day at a constant flow rate over the day. The daily volume of feed multiplied by the selected COD concentration gives the mass of COD fed per day.

Sludge age control: As with the RBCOD system, the sludge age is set by hydraulic control, e.g. for 10 and 20 days sludge age, every day 1/10th and 1/20th of the reactor volume respectively is abstracted. In time, the reactor MLVSS concentration will reach a steady state value corresponding to the set sludge age and the mass of COD fed per day.

Unlike the flow through activated sludge system for RBCOD determination, knowledge of the exact sludge age is essential in determining UPCOD. Accordingly, one needs to take care that additional losses of sludge are duly accounted for. In an experimental investigation usually the mass of VSS in the settling tank is small and the mass lost in the effluent is negligible. Under such conditions hydraulic control gives the correct sludge age. If the mass of sludge in the settling tank is found to be significant, this mass needs to be added to that in the reactor to give the ***total mass of VSS in the system.*** If the concentration of VSS in the effluent (measured directly or calculated from the difference between the unfiltered and filtered COD concentrations divided by the COD/VSS ratio) is significant, then the VSS mass lost via the effluent (concentration times effluent flow) needs to be added to the VSS mass abstracted from the reactor to give the ***total VSS mass wasted and lost from the system per day.*** From the total mass of VSS in the system and the total mass of VSS wasted and lost per day, a revised value of the sludge age can be calculated, i.e.

$$\text{Sludge age} = \frac{\text{total VSS mass in system}}{\text{total VSS mass wasted and lost per day}} \quad (d) \qquad (6.3)$$

To minimize wall growth in the system, which also will influence the sludge age, about every 2 weeks the mixed liquor must be emptied and stored under aeration while the reactor, settling tank and connecting tubing are thoroughly scrubbed in boiling water.

Reactor MLSS concentration and volume requirements: It is recommended that the experimental systems are operated with reactor mixed liquor suspended solids (MLSS) concentrations of approximately 4000 mgMLSS/ℓ ($\approx$ 3000 mgMLVSS/ℓ); higher concentrations may give rise to difficulties in sludge settling, aeration and mixed liquor recycling, while lower concentrations make reliable measurements difficult. The reactor volume is set to give the selected MLSS concentration; the reactor volume is determined by diluting the ***mass*** of MLSS expected for the sludge age and influent COD mass loading per day, to the selected MLSS concentration. From WRC (1984), to give the selected MLSS concentration of 4000 mgMLSS/ℓ at sludges of 10 and 20 days, the reactor volume requirements are approximately 0,9 and 1,5 ℓ/gCOD load respectively for raw wastewater and 0,4 and 0,6 ℓ/gCOD load respectively for settled wastewater.

As an example, if 10 ℓ/d of raw wastewater with a COD of 500 mg/ℓ is fed to the two experimental systems at 10 and 20 days sludge age then the required reactor volumes (V_p) are:

10 days: $V_p = 0{,}9 \cdot 10 \cdot 500/1000 = 4{,}5\ell$
20 days: $V_p = 1{,}5 \cdot 10 \cdot 500/1000 = 7{,}5\ell$

Parameter measurement
Once steady state has been achieved, usually after 2 to 3 sludge ages, the following parameters must be measured regularly over a further 2 to 3 sludge ages; filtered (0,45 μm) and unfiltered influent COD and TKN; filtered and unfiltered effluent COD and TKN; filtered effluent ammonia and nitrate; reactor MLSS and MLVSS; the COD/VSS and TKN/VSS ratios of the VSS and the oxygen utilization rate (OUR) in the reactor. The operation and control of these plants and the procedures for measuring the cited parameters, in particular the oxygen utilization rate, must be done with the utmost care. Practical details of these tests are set out at length by Marais and Ekama (1976).

Calculation of USCOD
From extensive experimental enquiry at the University of Cape Town, it became clear that for municipal-type wastes the filtered effluent COD is virtually constant

for all sludge ages longer than about 3 days. This was found to be so not only under constant flow and load conditions, but also under daily cyclic loading conditions. Furthermore, from measured oxygen utilization rates (OUR), at sludge ages greater than 5 days at 20°C, it was found that virtually all the biodegradable COD (both soluble and particulate) is utilized in the system; this behaviour has been reflected in numerous computer simulations of a wide range of experimental systems. Accordingly, for the experimental systems recommended above, operated at sludge ages of 10 to 20 days at 20°C, the effluent (soluble) biodegradable COD concentration can be assumed to be zero. Furthermore, the organisms in the experimental systems do not generate unbiodegradable COD to any significant degree (Dold *et al.*, 1986). Therefore, it can be accepted that the filtered effluent COD is due entirely to the USCOD in the influent and, in the two experimental systems, the USCOD is given by the COD of the filtered effluent (0,45μm filter). This COD concentration divided by the influent total COD (TCOD) gives the fractional constant $f_{S,us}$ required in the programs, i.e.

$$f_{S,us} = USCOD/TCOD = S_{usi}/S_{ti} \qquad (6.4)$$

Estimation of UPCOD
The influent UPCOD is estimated by comparing MLVSS concentrations measured for the two experimental systems, with MLVSS concentrations predicted by the two models, as follows:

The experimental system set-ups serve as input to the program. Simulations are run for a range of $f_{S,up}$ values; the $f_{S,up}$ value which gives the predicted MLVSS closest to that measured at both sludge ages is the estimated $f_{S,up}$ for that wastewater.

Provided that good COD and N mass balances (for determination see later) are obtained (95 to 105%) for the experimental systems, the $f_{S,up}$ determined above can be checked as follows:

Compare the ***predicted*** oxygen utilization rate (OUR) for the carbonaceous material against the ***measured*** OUR for carbonaceous material. If nitrification takes place, the measured OUR must be corrected to give the OUR for carbonaceous material only by deducting the OUR for nitrification. Provided a good N balance is obtained (>95%, i.e. no inadvertent denitrification in system) the OUR for nitrification (OUR_n) can be calculated from the measured effluent nitrate concentration (N_{o3e}) less the influent nitrate concentration (N_{o3i}), i.e.

$$OUR_n = 4{,}57 \cdot (N_{o3e} - N_{o3i}) \cdot Q/(24 \cdot V_p) \qquad (mgO/\ell/h) \qquad (6.5)$$

where
4,57 = mg oxygen required to nitrify 1 mgN nitrate (mgO/mgN)
V_p = reactor liquid volume (ℓ)

[In the above equation (6.5) nitrite is not included; in completely aerobic systems the nitrite concentration usually is small and can be neglected.]

Hence, the measured OUR for carbonaceous material (OUR_c) is given by

$$OUR_c = OUR_m - OUR_n \qquad (mgO/\ell/h) \qquad (6.6)$$

where
OUR_m = measured OUR (mgO/ℓ/h)

These calculations only apply if there is no denitrification in the experimental systems. Also, the OUR is relatively insensitive to variations in $f_{S,up}$ so that accurate and reliable measurements are essential; these can be checked by COD and N mass balances, see below.

In utilizing the procedure above to determine $f_{S,up}$, the values of the kinetic/stoichiometric constants Y_{ZH}, b_H, f_E and f_{cv} can influence the predicted MLVSS and hence the estimated value of $f_{S,up}$; values for these constants therefore must be known. The default values for these constants supplied in the programs can be accepted as adequate; these values have been obtained from extensive research enquiry. Should the values for the constants require calibration, the reader is referred to Marais and Ekama (1976), WRC (1984) and Ekama *et al.* (1986).

Mass balances

The accuracy of the values for UPCOD ($f_{S,up}$) determined via the procedures set out above hinges around the acceptable accuracy of the experimental data. Acceptability is checked by calculating N and COD mass balances over the experimental system. In the N balance, the influent TKN must be accounted for (to within 95 to 105%) by the sum of the mass of effluent TKN and nitrate, and the mass of N abstracted with the waste sludge. Similarly, the influent COD must be accounted for (to within 95 to 105%) by the sum of the masses of effluent COD, carbonaceous oxygen demand and COD of the waste sludge. For the two experimental systems described earlier, the methods for calculating N and COD mass balances are given below:

N balance:

(1) Mass of N in the sludge wasted per day (ΔMN_X) is given by the TKN/VSS ratio times the mass of VSS wasted from the reactor per day, i.e.

$$\Delta MN_X = f_{X,N} \cdot q \cdot X_v = f_{X,N} \cdot V_p \cdot X_v / R_s \quad \text{(mgN/d)} \tag{6.7}$$

where

q = wastage flow rate (ℓ/d)
X_v = reactor VSS concentration (mgVSS/ℓ)
V_p = reactor liquid volume (ℓ)
R_s = sludge age (d)
$f_{X,N}$ = TKN/VSS (mgN/mgVSS)
= fraction of VSS which is N ≈ 0,1 mgN/mgVSS.

(2) Mass of TKN-N in the effluent (MN_{te}) is given by

$$MN_{te} = Q \cdot N_{te} \quad \text{(mgN/d)} \tag{6.8}$$

where

Q = influent flow rate (ℓ/d)
N_{te} = effluent TKN concentration (mgN/ℓ)

In Eq (6.8), the influent flow rate, not the effluent flow rate, is used. The effluent flow rate is less than the influent flow rate by the sludge waste flow rate; soluble TKN equal in concentration to the effluent TKN also leaves the system via the sludge waste flow. Using the influent flow rate takes this into account. Also, in Eq (6.8), if N_{te} is the filtered value, then in Eq (6.7) the value for the sludge age must be adjusted to take due account of the VSS in the effluent, see Eq (6.3). Alternatively, if N_{te} is the unfiltered value then the value for the sludge age need not be adjusted.

(3) Mass of nitrate generated in the system (MN_{o3}) is given by

$$MN_{o3} = Q \cdot N_{o3e} - Q \cdot N_{o3i} \qquad (mgN/d) \qquad (6.9)$$

where
N_{o3e} = effluent nitrate concentration (mgN/ℓ)
N_{o3i} = influent nitrate concentration (mgN/ℓ)

From Eqs (6.7 to 6.9) the N balance is given by

$$N_{bal}\,(\%) = 100 \cdot (\Delta MN_X + MN_{te} + MN_{o3})/MN_{ti} \qquad (6.10)$$

where
MN_{ti} = influent TKN mass (mgN/d)
= $Q \cdot N_{ti}$ (6.11)
N_{ti} = influent TKN concentration (mgN/ℓ)

The experimental data can be considered acceptable if the N balance is in the range 95 to 105%. If the N balance does not fall within this range, the source of error should be identified in the experimental system, sample handling or analysis techniques, and the test repeated.

COD balance:
(1) The COD of the sludge mass wasted per day ($MCOD_w$) is given by the COD/VSS ratio (f_{cv}) times the mass of VSS wasted per day, i.e.

$$MCOD_w = f_{cv} \cdot q \cdot X_v = f_{cv} \cdot V_p \cdot X_v/R_s \qquad (mgCOD/d) \qquad (6.12)$$

where
f_{cv} = COD/VSS ratio of the sludge ≈ 1,48 mgCOD/mgVSS.

(2) The COD mass in the effluent (MS_{te}) is given by

$$MS_{te} = Q \cdot S_{te} \qquad (mgCOD/d) \qquad (6.13)$$

where
Q = influent flow rate (ℓ/d)
S_{te} = effluent COD concentration (mgCOD/ℓ).

In Eq (6.13), if S_{te} is the filtered value, then in Eq (6.12) the value for the sludge age must be adjusted to take due account of the VSS in the effluent, see Eq (6.3). Alternatively, if S_{te} is the unfiltered value then the sludge age need not be adjusted.

(3) The COD recovered from the oxygen demand must take due account of the oxygen demand for nitrification. Provided the N balance is acceptable, the oxygen demand for nitrification (MO_n) is given by

$$MO_n = 4{,}57 \cdot MN_{o3} \qquad (mgO/d) \qquad (6.14)$$

where
MN_{o3} = mass of nitrate generated per day, Eq (6.9) (mgN/d)
4,57 = mass of oxygen utilized per ammonia nitrified (mgO/mgN).

From the measured OUR (mgO/ℓ/h) the oxygen demand for carbonaceous material (MO_c) is given by

$$MO_c = (OUR)_M \cdot V_p \cdot 24 - MO_n \quad (mgO/d) \tag{6.15}$$

where
$(OUR)_M$ = measured OUR (mgO/ℓ/h).

From Eqs (6.12 to 6.15), the COD balance is given by

$$COD_{bal}\,(\%) = 100 \cdot (MCOD_w + MS_{te} + MO_c)/MS_{ti} \tag{6.16}$$

where

$$\begin{aligned} MS_{ti} &= \text{total mass of COD fed to system (mgCOD/d)} \\ &= Q \cdot S_{ti} \\ S_{ti} &= \text{influent COD concentration (mgCOD/ℓ).} \end{aligned} \tag{6.17}$$

As with the N balance, the COD balance also should be at least 95 to 105% for the experimental data to be acceptable for aerobic systems[2]. From Eq (6.15) it can be seen that errors in the N balance are carried through to the COD balance via the oxygen demand for nitrification. Consequently both N and COD balances should be adequate for the data to be acceptable. Generally, greater difficulty is found in achieving good COD balances than N balances: Likely sources of error in the COD balance arise from loss of COD in the feed bucket before discharge to the system, incorrect determination of the OUR and/or errors in the measurement of the influent COD; a likely source of error in the N balance arises in aerobic systems from unintentional denitrification in the experimental set-up.

In the calculation of the mass balances only 3 stoichiometric constants are required, *viz* COD/VSS ratio (f_{cv}), TKN/VSS ratio ($f_{X,N}$) and 4,57 mgO required per $mgNO_3$-N generated by nitrification. The first two of these constants can be directly measured in the investigation. The latter is a theoretical value but it has been experimentally verified and shown to be sufficiently accurate for all purposes. Consequently in the mass balances a minimum of theoretical assumptions are applied – the data are checked on the basis of continuity and conservation of mass principles which do not invoke theoretical relationships depicting the activated sludge system kinetics. Only with acceptable mass balances are the data adequate to calculate reliably kinetic and sewage characteristic constants within the framework of the theoretical model. ***To obtain acceptable N and COD mass balances, the experimental investigation must be conducted with an uncompromising vigilance, a strict discipline and attention to detail in every operational procedure; if these are lax or neglected, the results will be largely useless.***

INFLUENT TKN CHARACTERISTICS

Knowledge of influent TKN characteristics is of importance in determining nitrification oxygen demand, nitrification capacity and effluent TKN. As with the COD, the influent TKN also is subdivided into different fractions. In **Chapter 2** it was shown that the TKN of wastewaters can be divided in two main fractions (Fig 6.4), (1) free and saline ammonia and organically bound N. The organically bound

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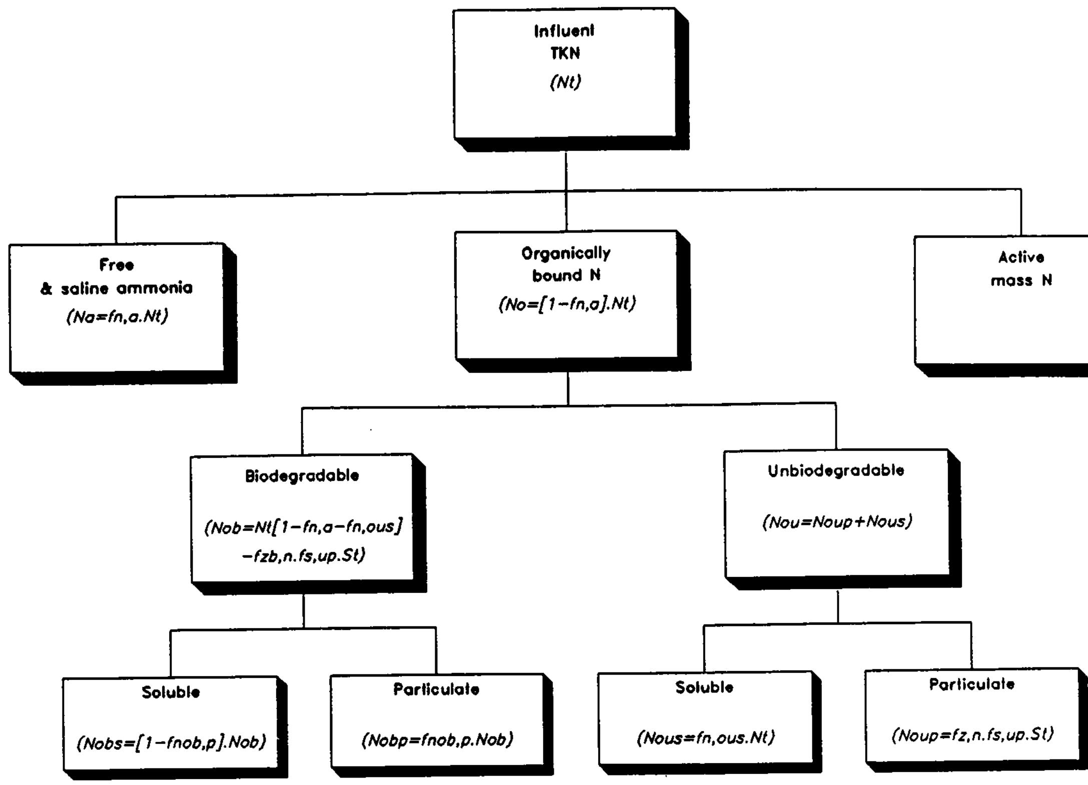
[2]Difficulties may be encountered in obtaining good COD balances in systems with large unaerated mass fractions (Arkley and Marais, 1981; Warburton *et al.*, 1991). In contrast, good N balances have been obtained consistently in all types of systems.

N has two subfractions, biodegradable and unbiodegradable; each of these is further subdivided into particulate and soluble subfractions to give (2) biodegradable particulate organic N (BPON), (3) biodegradable soluble organic N (BSON), (4) unbiodegradable particulate organic N (UPON) and (5) unbiodegradable soluble organic N (USON). Following the IAWPRC recommendations, an influent biological (active) mass N fraction is included in both models, see Fig 6.4. However, quantification of this fraction is via the influent biological (active) mass COD fraction which can be determined only by trial and error fitting of simulated to experimental data (see **Chapter 2**). Accordingly, procedures to determine this N fraction are not included in the discussion below.



<u>Fig 6.4</u>: Division of the influent TKN into its constituent fractions.

With regard to quantification of the other influent N fractions, the free and saline ammonia (N_{ai}) is measured directly by the test bearing this name; the total organic N_{oi} is found from the difference between the measured influent TKN (N_{ti}) and free and saline ammonia (N_{ai}) concentrations. Difficulties in quantification arise when subdividing the total organic N (N_{oi}) into its four subfractions, BPON, BSON, UPON and USON (see Fig 6.4). Accurate measures of the magnitudes of these fractions can be obtained only by comparing the observed response of laboratory-scale systems (such as those described in the previous section '**<u>Determination of unbiodegradable soluble and particulate COD</u>**') with that predicted by the two models. From a large number of such comparisons for systems operated within the range of sludge ages 2 to 30d under cyclic and steady state in the temperature range 12 to 25°C, it would appear that subdivision of the total organic N into four subfractions is necessary in the models to obtain consistent correlation between observation and prediction. However, the magnitudes of the two unbiodegradable organic N fractions, UPON and USON, are small and the relative subdivision of the biodegradable organic N into the two subfractions, BPON and BSON, does not exert significant influence on the model's predictions. It is not necessary therefore to have accurate measures for the magnitudes of UPON and USON or for the subdivision of the biodegradable organic N into soluble (BSON) and particulate (BPON) fractions. Accordingly, for most applications the default values for the constants defining the total organic N fractions ($f_{N,ous}$, $f_{Nob,p}$) can be accepted as adequate. However, should it prove necessary, the magnitudes of the total organic N fractions can be estimated from measurements made on the laboratory-scale systems described in the previous section '<u>Determination of unbiodegradable soluble and particulate COD</u>', provided acceptable COD and N balances have been obtained.

UPON

In the models UPON in the influent (N_{oupi}) is expressed in terms of UPCOD in the influent (S_{upi}), i.e.

$$N_{oupi} = f_{Z,N} \cdot S_{upi} \qquad (6.18)$$

where

$f_{Z,N}$	= TKN/COD ratio of MLVSS	(mgN/mgCOD)
	$= f_{cv} \cdot f_{X,N}$	
$f_{X,N}$	= TKN/VSS ratio of MLVSS	(mgN/mgVSS)
	= ≈ 0,1 mgN/mgVSS	
f_{cv}	= COD/VSS ratio of MLVSS	(mgCOD/mgVSS)
	= ≈ 1,48 mgCOD/mgVSS	

From experimental data on the TKN/COD ratio of the mixed liquor volatile suspended solids (MLVSS) in the biological reactor, our experience is that the TKN/COD ratio remains substantially the same irrespective of the sludge age from 3 to 30d despite the fact that the various MLVSS components (active, endogenous and inert fractions) change with sludge age. Hence, it would seem reasonable to accept that the TKN/COD ratio of the inert fraction of the MLVSS (and therefore of UPCOD) is equal to that of the other MLVSS fractions. The TKN/COD ratio for MLVSS, from extensive experimental measurements has a mean value of 0,068 mgN/mgCOD. The value for the TKN/COD ratio in a particular investigation can be checked by direct measurement of the TKN and COD of the MLVSS in the two experimental systems. If necessary, the computer program input constants for the TKN/COD ratio, $f_{ZB,N}$ and $f_{ZE,N}$, can be adjusted to reflect the measured TKN/COD ratio.

USON

An estimate of USON in the influent (N_{ousi}) can be obtained from the filtered (<0,45μm) effluent total TKN of the two experimental systems. The filtered effluent total TKN (N_{te}) contains three soluble N fractions, USON, BSON and free and saline ammonia. The effluent free and saline ammonia (N_{ae}) can be measured directly by the test bearing this name. (The N_{ae} very likely will be small so that reliable measurement of this parameter may not be possible; if $N_{ae} < 1,0$ mgN/ℓ then accept $N_{ae} = 0,5$ mgN/ℓ). From numerous simulations, for sludge ages > 5 days and temperature = 20°C the effluent BSON (N_{obse}) can be accepted to be 1,5 mgN/ℓ. Accordingly, the effluent USON (N_{ouse}) is given by

$$N_{ouse} = N_{te} - N_{ae} - 1,5 \qquad (6.19)$$

Since USON is unbiodegradable and soluble, the influent concentration will equal the effluent concentration, i.e.

$$N_{ousi} = N_{ouse} \tag{6.20}$$

As input to the computer programs, the fractional constant $f_{N,ous}$ defines the influent USON where

$$f_{N,ous} = N_{ousi}/N_{ti} \tag{6.21}$$

BSON

An estimate of BSON in the influent (N_{obsi}) can be obtained from the filtered (<0,45μm) ***influent*** TKN. The filtered influent TKN contains three soluble N fractions, free and saline ammonia, USON and BSON. The influent free and saline ammonia (N_{ai}) can be measured by the test bearing this name and the influent USON (N_{ousi}) has been estimated in the previous section. The filtered influent TKN remaining can be assigned to the influent BSON (N_{obsi}) fraction, i.e.

$$N_{obsi} = \text{filtered influent TKN} - N_{ai} - N_{ousi} \tag{6.22}$$

The input required for the computer programs is the fractional constant $f_{Nob,p}$ which defines the influent BSON in terms of the influent biodegradable organic N (N_{obi}), i.e.

$$N_{obi} = N_{ti} - N_{ai} - N_{ousi} - N_{oupi} \tag{6.23}$$

and

$$f_{Nob,p} = 1 - N_{obsi}/N_{obi}$$

BPON

The influent BPON (N_{obpi}) is given by the influent TKN remaining after all the other TKN fractions have been estimated, i.e.

$$N_{obsi} = N_{ti} - N_{ai} - N_{obsi} - N_{ousi} - N_{obsi} \tag{6.24}$$

N_{obpi} is not required as input to the computer programs, but is calculated automatically by the programs from the influent TKN fractions given.

Revised estimates for organic N fractions

In the procedures described above to estimate the various organic N fractions, initially it was assumed that the effluent BSON (N_{obsi}) was 1,5 mgN/ℓ, see Eq (6.19). A revised estimate for this parameter can be obtained by inputting the initial estimates for the various TKN fractions (as determined above) into the computer programs and simulating the two experimental systems. The predicted value for the effluent BSON is accepted, and the various organic N fractions recalculated via the procedures detailed above. This process is repeated until the predicted effluent BSON is the same for two successive simulations.

KINETIC AND STOICHIOMETRIC CONSTANTS

In **Chapter 2**, the kinetic and stoichiometric constants for the two models are listed together with the default values included in the programs. These default values have been obtained from extensive research enquiry, and have been consistently corroborated in application of the models to simulate data obtained from a range of experimental systems (see **Chapter 5** for examples). Therefore, we believe the quoted default values constitute reliable average values for normal application of the models to systems treating municipal-type wastes. ***However, it should be noted that these values are for temperature = 20° C and pH = 7,2 to 8,5.*** Adjustment of the values for the operational temperature of the activated sludge system to be simulated is computed automatically by the programs from the input temperature and the default Arrhenius temperature constants; adjustment for pH is not computed (refer WRC, 1984 for details on adjusting the values for pH). Furthermore, from experience it has been found that the values for some of the constants change for different wastewaters and/or system configurations and load patterns. In this regard four constants have been found to change:

(1) The maximum specific growth rate of the heterotrophs on RBCOD ($\hat{\mu}_H$) is dependent on the system configuration and load pattern (Ekama *et al.*, 1986, Gabb *et al.*, 1989). For most applications of the models the value for this constant has little influence on the models predictions - the rate of growth on RBCOD usually is limited by the rate at which RBCOD enters the system with the influent and not by the kinetics of utilization. However, the value has significance in that it sets the denitrification rate for RBCOD in *batch* denitrification which forms the basis in design to fix the minimum size for a primary anoxic reactor (WRC, 1984). Also, μ_H is implicated in the phenomenon of sludge bulking by certain filamentous organisms.[3]

(2) The maximum specific growth rate of the autotrophs ($\hat{\mu}_A$) is dependant on the wastewater (WRC, 1984) and on the system configuration and load pattern (Still *et al.*, 1986). Again, for most applications of the models the absolute value for this constant has little influence on the models predictions provided the value selected falls in the range 0,2–0,8/d and the system nitrifies completely at the lowest temperature. However, the value for this constant has significance in design in that it influences the minimum sludge age for complete nitrification (WRC, 1984).

(3) Change in the value of the half saturation constant for heterotrophic RBCOD utilization (K_{SH}) has been found necessary in simulations of the contact-stabilization system, from K_{SH} = 5 to 110 mgCOD/ℓ for **UCTOLD** and from K_{SH} = 5 to 50 mgCOD/ℓ for **IAWPRC** (see **Chapter 5** for details). However, whether this change is a true reflection of the behaviour of the organisms in a contact-stabilization system is not certain. In all other situations investigated, the default value for K_{SH} was found to be adequate, so that an independent determination of this constant is not necessary.

[3]In investigations into sludge bulking at the University of Cape Town, it has been found that the value for $\hat{\mu}_H$ has no significance in bulking by low F/M filaments, but may be of importance for bulking due to other filamentous organisms (Gabb *et al.*, 1991).

(4) Change in the value for the SBCOD hydrolysis rate half saturation constant (K_{SP} for **UCTOLD** and K_X for **IAWPRC**) has been found necessary in simulations of batch tests in which nitrification has been inhibited by addition of thiourea (see **Chapter 5**). The constant needed to be changed from 0,027 to 0,15 mgCOD/mgCOD for both models. However, in all other situations investigated, the default value of 0,027 mgCOD/mgCOD was found to be adequate, so that independent determination of this constant probably is not necessary.

From the above, variability in $\hat{\mu}_H$ and $\hat{\mu}_A$ indicate that these two constants should be determined before model application if a sensitivity analysis shows that the values for the constants have kinetic influence, i.e. if variation in $\hat{\mu}_H$ of 1 to 6/d and $\hat{\mu}_A$ of 0,2 to 0,8/d significantly influence the model predictions.

<u>Measurement of $\hat{\mu}_H$ and $\hat{\mu}_A$</u>

The constants $\hat{\mu}_H$ and $\hat{\mu}_A$ are quantified by means of aerobic batch tests. In this test a volume of wastewater is mixed with a volume of mixed liquor in an aerated and stirred batch reactor, and the OUR and nitrate responses monitored with time. Three important aspects must be noted in obtaining wastewater and mixed liquor for the batch test:

(1) Because $\hat{\mu}_A$ is dependant on the wastewater, the wastewater added to the batch test must be the same as that fed to the system (to be simulated using the computer programs).

(2) Because $\hat{\mu}_H$ and $\hat{\mu}_A$ are dependant on the system configuration and load pattern, the mixed liquor added to the batch test must be drawn from the system to be simulated, or from a pilot-scale/laboratory-scale replica of the system; it is not possible to use mixed liquor drawn from a different laboratory-scale system (such as those detailed in the previous section) because the values for $\hat{\mu}_H$ and $\hat{\mu}_A$ may differ from those in the system to be simulated. Also, using mixed liquor and wastewater from the same system will overcome acclimatization problems. (If a pilot-scale/laboratory-scale replica of the system is operated, it must be fed the same wastewater as the system.)

(3) The ***mixed liquor must not be drawn from a biological excess phosphorus removal system*** – in the aerobic batch test, on addition of the wastewater, the organisms mediating biological excess phosphorus removal (polyP organisms) will take up some of the RBCOD and store it as PHB (Wentzel *et al.*, 1989), and in this manner interfere with the response of the heterotrophs.

Basis of Test
With the correct selection of the volumes of wastewater and mixed liquor (see later), on mixing in the aerated, stirred batch reactor, the OUR remains constant at a plateau for a period of up to 3 hours depending on the wastewater RBCOD fraction, whereafter the OUR drops precipitously, then levels off at a second plateau, see Fig 6.5. This OUR–time profile is made up of the OUR's for nitrification and for carbonaceous material utilization.

If the ammonia concentration is greater than about 2 times the half saturation constant for the nitrification (in the Monod formulation for the growth rate of the nitrifying autotrophs, **Chapter 2**), the nitrification will take place at a maximum (and constant) rate. Consequently, if adequate ammonia is available at the start of the batch test, and since the half saturation constant for the autotrophs is relatively small (K_{SA} = 1,0 mgN/ℓ), the nitrification *rate* will remain at a maximum and constant value over the test period giving rise to a constant nitrification OUR (Fig 6.5, Area 3) and a linear increase in the nitrate concentration (Fig 6.6). Since the nitrification rate is at a maximum and approximately constant, $\hat{\mu}_A$ can be determined from the nitrate concentration profile. If inadequate ammonia-N is present, the observed OUR will exhibit a second drop to a third plateau level and the nitrate concentration profile will reflect this decrease in nitrification rate – in this case it is necessary to redo the test supplementing the wastewater with ammonia-N.

Superimposed on the constant nitrification OUR is the OUR due to carbonaceous material utilization [Fig 6.5, Area 1 + Area 2(a and b)] which gives rise to the variation in the observed OUR with time. The interpretation of the carbonaceous OUR-time profile differs in terms of the UCT and IAWPRC Task Group models:

UCT model: In the UCT model, RBCOD and SBCOD are utilized independently but simultaneously by the heterotrophs for growth; RBCOD is directly absorbed and utilized while SBCOD is adsorbed extra cellularly, hydrolysed to smaller units which then are utilized directly (**Chapter 2**). The summation of the OUR's associated with RBCOD and SBCOD growth gives rise to the observed carbonaceous OUR. At the start of the batch test both RBCOD and SBCOD are present, and are utilized independently for growth, with associated OUR's (Fig 6.5, Area 1 for RBCOD, Area 2a for SBCOD). The OUR is constant over the initial period because the RBCOD and SBCOD concentrations are sufficiently high to ensure that RBCOD utilization and SBCOD hydrolysis/utilization rates are close to their respective maxima. Once the influent RBCOD is depleted, the OUR drops to the second plateau which is due to the maximum hydrolysis/utilization of SBCOD only (Fig 6.5, Area 2b). The magnitude of the drop in OUR is proportional to the heterotroph maximum specific growth rate on RBCOD, $\hat{\mu}_H$, and can be used to obtain an estimate for this constant (Fig 6.5). Also, the area under the OUR-time plot associated with RBCOD (Fig 6.5, Area 1) can be used to calculate the wastewater RBCOD concentration.

IAWPRC Task Group model: In the IAWPRC Task Group model only RBCOD is utilized for growth, the SBCOD is hydrolysed to RBCOD (**Chapter 2**). Therefore, the carbonaceous OUR is due only to that OUR associated with RBCOD utilization. In the batch test the initial high OUR is a consequence of the utilization of RBCOD derived from the influent (Fig 6.5, Area 1) and from hydrolysis of SBCOD (Fig 6.5, Area 2a). The OUR is constant over this period because the concentration of RBCOD is so high that the growth rate of the heterotrophs is at a maximum ($\hat{\mu}_H$) in accordance with the Monod kinetics for growth on RBCOD (see **Chapter 2**). Because of the saturation condition it is possible to estimate $\hat{\mu}_H$ from the initial high OUR (Fig 6.5) and to estimate the influent RBCOD concentration from the area under the high OUR plateau (Fig 6.5, Area 1). Once the RBCOD in the influent is depleted, the OUR drops to the second plateau level, which is associated with the utilization of RBCOD generated from hydrolysis of SBCOD (Fig 6.5, Area 2b).

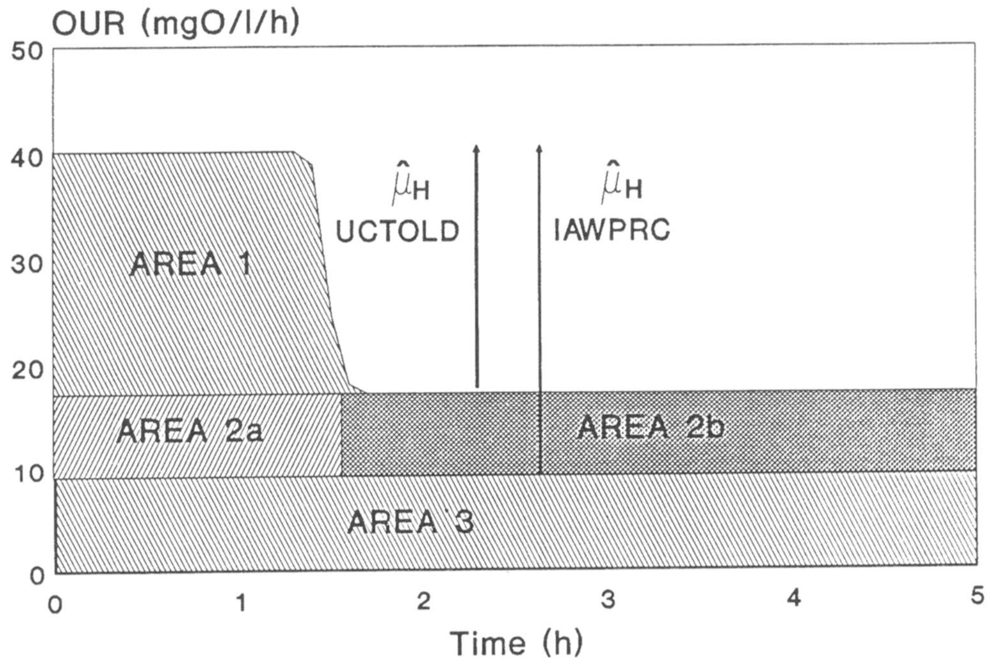


Fig 6.5: OUR-time profile for aerobic batch test with nitrification; see Fig 6.6 for nitrate-time profile.

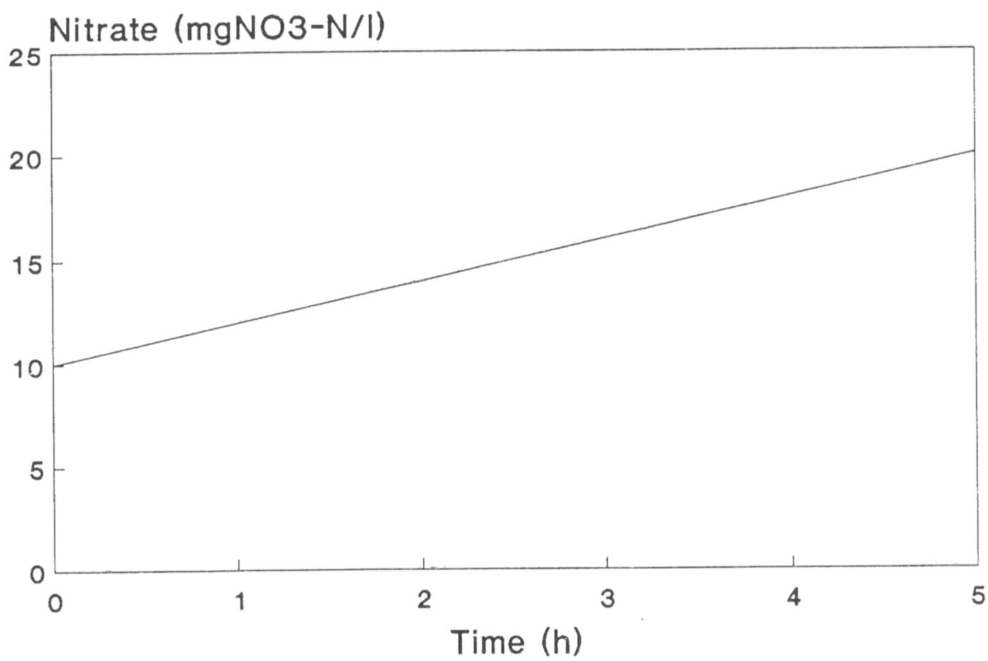
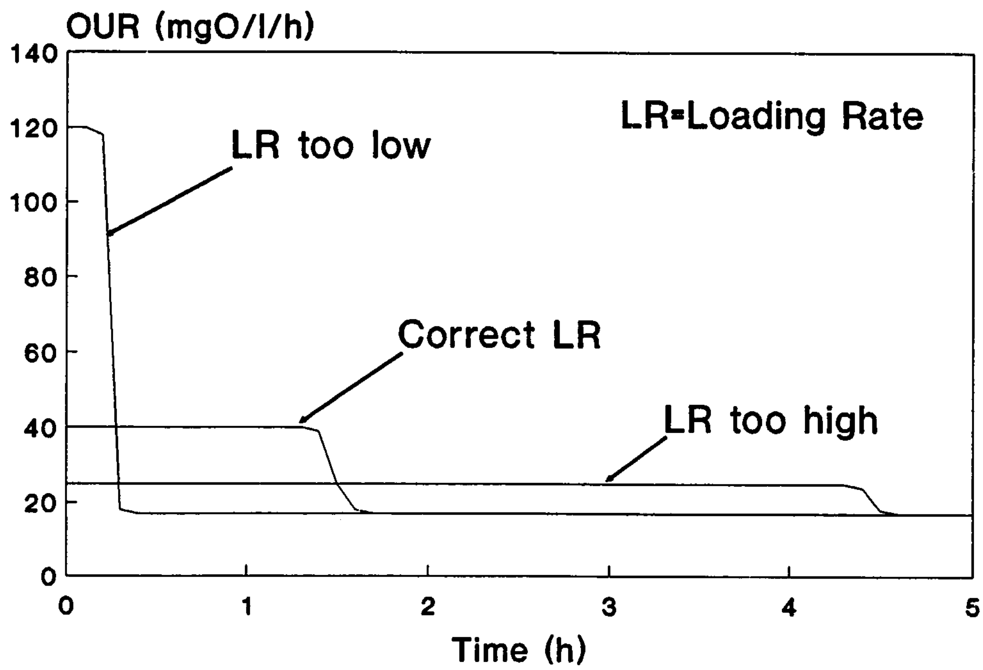


Fig 6.6: Nitrate-time profile for aerobic batch test with nitrification; see Fig 6.5 for OUR-time profile.

Aerobic batch test method

Taking note of the considerations above, a preselected volume of mixed liquor (V_{ml}) of known MLVSS concentration (X_v) is placed in an aerated, stirred batch reactor. A preselected volume of wastewater (V_{ww}) of known characteristics (for procedures to characterize the wastewater see earlier) is added. From immediately after mixing, the OUR is measured approximately every 5 to 10 minutes for at least about 4 to 5 hours using the technique described previously for **RBCOD measurement**. Also, periodically samples are extracted from the batch reactor, biological activity terminated immediately by adding a drop of $HgCl_2$ solution (8,6g $HgCl_2/\ell$), the samples filtered (0,45μm) and nitrate measurements made on the filtrate. The temperature and pH in the batch reactor must be monitered; pH must be maintained in the region 7,2 to 7,5 and temperature (T°C) kept constant in the range 14 to 22°C. In selection of V_{ww} and V_{ml} the following warrants attention:

> *COD loading rate:* The mass of COD (i.e. $V_{ww}S_{ti}$ where S_{ti} = total COD concentration of the undiluted wastewater) with respect to the mass of VSS (i.e. $V_{ml}X_v$) mixed in the batch test is known as the COD loading rate (LR). The LR established in the batch test should be such that the OUR response is well defined and allows (1) the initial peak OUR to be readily determined (to estimate $\hat{\mu}_H$ in the IAWPRC Task Group model), (2) the magnitude of the precipitous drop in OUR to be clear (to estimate $\hat{\mu}_H$ in the UCT model) and (3) the area under the initial peak OUR (Fig 6.5, Area 1) to be accurately estimated (to calculate RBCOD). For the ***same wastewater volume*** changing the LR does not change the ***magnitude*** of Area 1, which is a function only of the mass of RBCOD in the wastewater sample, but it does change the ***shape*** of Area 1: If LR is too low (i.e. a higher mixed liquor volume), the shape is tall and narrow because the RBCOD is utilized very quickly, and too few OUR measurements can be taken to give reasonable surety of the initial high OUR; if LR is too high (i.e. a lower mixed liquor volume), the shape is low and wide and it is difficult to establish the magnitude of the drop in OUR (see Fig 6.7).



<u>Fig 6.7</u>: OUR-time profile for aerobic batch test to show the effect of loading rate on the shape of the OUR-time response.

Ideally the step change in OUR should take place 1 to 2 hours after the start of the test. This can be achieved by setting LR at 1,0 to 1,2 times the estimated heterotroph (active) biological mass fraction of the VSS, $f_{X,BH}$, i.e.

$$LR = (1{,}0 \text{ to } 1{,}2) \cdot f_{X,BH} \qquad \text{(mgCOD/mgVSS)} \qquad (6.25)$$

The value for $f_{X,BH}$ is a function of the sludge age (R_s) of the system from which the mixed liquor is obtained and can be ***estimated roughly*** as follows (Ekama *et al.*, 1986):

$$f_{X,BH} = 1{,}41 \cdot (R_s)^{-0{,}53} \text{ for raw wastewater} \qquad (6.26a)$$

and

$$f_{X,BH} = 1{,}57 \cdot (R_s)^{-0{,}43} \text{ for settled wastewater} \qquad (6.26b)$$

(Although the two equations above provide a rough estimate for $f_{X,BH}$, they are not sufficiently accurate to be used to estimate $\hat{\mu}_H$, as described later).

Selection of the appropriate LR is complicated by the fact that $\hat{\mu}_H$ varies for different system configurations and load patterns (Still *et al.*, 1986; Gabb *et al.*, 1989). Variation in $\hat{\mu}_H$ will result in different OUR-time response profiles for different mixed liquors even though the LR is kept constant. The equations above, therefore, provide only an initial estimate for the correct LR, the best value for this ratio can be determined only by trial and error.

Note that if the LR in a batch test is 'excessively' high the kinetic response of the organisms deviates from the pattern described above. The maximum OUR during the period when RBCOD is present increases with time until the RBCOD is depleted giving a 'saw tooth' appearance instead of a flat plateau.

Maximum OUR: The maximum OUR in the test should be at a value that can be measured conveniently and accurately, i.e. at values which allow the DO concentration to be raised quickly. From experience, a convenient OUR is about 30 to 40 mgO/ℓ/h. The OUR is governed by the active mass concentration in the batch test, which in turn is related to the heterotroph biological mass fraction ($f_{X,BH}$) and the MLVSS concentration. The MLVSS concentration in the batch test (X_vBT), to give an OUR of 30 to 40 mgO/ℓ/h, can be estimated from

$$X_vBT = (300 \text{ to } 400)/f_{X,BH} \qquad (\text{mgVSS}/\ell \text{ batch volume}) \qquad (6.27)$$

where the value for $f_{X,BH}$ can be estimated from Eq (6.26).

It may not be possible to attain the MLVSS in the batch test in this range especially when the wastewater COD concentration is low; for these feed concentrations, lower MLVSS concentrations will be obtained in which event OUR's lower than 30 mgO/ℓ/h will be measured.

Calculation procedure

For estimation of $\hat{\mu}_A$, $\hat{\mu}_H$ and RBCOD, the MLVSS fractions of the parent system mixed liquor need to be determined.

Parent system MLVSS fractions: From the steady state theory (WRC, 1984) the various MLVSS fractions can be calculated as follows:[4]

- Heterotroph biological (active) mass

$$MX_{BH} = \frac{MS_{ti} \cdot (1 - f_{S,us} - f_{S,up}) \cdot Y_{XH} \cdot R_s}{(1 + b^*_{HT} \cdot R_s)} \tag{6.28a}$$

where

MX_{BH} = Heterotroph biological (active) mass in system (mgAVSS)
$= V \cdot X_{BH}$ (6.28b)
V = System volume (ℓ)
X_{BH} = Heterotroph biological (active) mass concentration (mgAVSS/ℓ)
MS_{ti} = Mass of COD load per day (mgCOD/d)
$= Q \cdot S_{ti}$ (6.28c)
Q = Influent flow rate (ℓ/d)
S_{ti} = Influent COD concentration (mgCOD/ℓ)
$f_{S,us}$ = Fraction of influent COD which is unbiodegradable soluble
$f_{S,up}$ = Fraction of influent COD which is unbiodegradable particulate
Y_{XH} = Heterotroph specific yield in VSS units (0,45 mgAVSS/mgCOD)
R_s = Sludge age (d)
b^*_{HT} = specific heterotroph endogenous mass loss rate[5] at temperature T° C
$= 0{,}24 \cdot (1{,}029)^{(T-20)}$ (/d) (6.28d)

- Endogenous mass

$$MX_E = f^*_E \cdot b^*_{HT} \cdot R_s \cdot MX_{BH} \tag{6.29a}$$

where

MX_E = Endogenous mass in system (mgEVSS)
$= V \cdot X_E$ (6.29b)
X_E = Endogenous mass concentration (mgEVSS/ℓ)
f^*_E = Fraction of biological (active) mass which is unbiodegradable particulate according to endogenous mass loss approach (0,2 mgEVSS/mgAVSS)

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[4]To regularize the symbol system the symbols used in this manual are not exactly the same as those used in WRC (1984); the reader is referred to **Appendix D** for details on the symbol system. Also, in the models described in **Chapter 2**, MLVSS is expressed in terms of COD units. In this Chapter MLVSS is expressed directly as VSS as this parameter is more practical to measure. To differentiate between these units, the symbol Z is used for COD units and the symbol X for VSS units, where $Z = f_{cv}X$ and f_{cv} = COD/VSS ratio of the MLVSS.

[5]The specific endogenous mass loss rate (b^*_H) used in this Chapter is not the same as the specific death rate (b_H) used in the models (see **Chapter 2**). However, under steady state conditions, applying the death-regeneration approach will give nett active mass loss identical to the endogenous mass loss approach. The endogenous mass loss approach is used here in preference to the death-regeneration approach because the calculation procedures are simpler.

- Inert mass

$$MX_I = f_{S,up} \cdot MS_{ti} \cdot R_s / f_{cv} \tag{6.30a}$$

where
MX_I = Inert mass in system (mgIVSS)
$= V \cdot X_I$ (6.30b)
X_I = Inert mass concentration (mgIVSS/ℓ)
f_{cv} = COD/VSS ratio of MLVSS (1,48 mgCOD/mgVSS)

- MLVSS

$$MX_v = MX_{BH} + MX_E + MX_I \tag{6.31a}$$

where
MX_v = mass of volatile suspended solids in system (mgVSS)
$= V \cdot X_v$ (6.31b)
X_v = Volatile suspended solids concentration (mgVSS/ℓ)

- Ratio heterotroph biological (active) mass/MLVSS

$$f_{X,BH} = MX_{BH}/MX_v \tag{6.32}$$

- Autotroph biological (active) mass

$$MX_{BA} = \frac{MN_{O3} \cdot Y_{XA} \cdot R_s}{(1 + b_{AT} \cdot R_s)} \tag{6.33a}$$

where
MX_{BA} = Autotroph biological (active) mass (mgVSS)
$= V \cdot X_{BA}$ (6.33b)
X_{BA} = Autotroph biological (active) mass concentration (mgVSS/ℓ)
MN_{O3} = Mass of nitrate generated per day (mgN/d)
$= MN_{ti} - MN_{te} - \Delta MN_x$ (6.33c)
MN_{ti} = Influent TKN mass load (mgN/d)
$= Q \cdot N_{ti}$ (6.33d)
N_{ti} = Influent TKN concentration (mgN/ℓ)
MN_{te} = Effluent TKN mass (mgN/d)
$= Q \cdot N_{te}$ (6.33e)
N_{te} = Effluent TKN concentration (mgN/ℓ)
ΔMN_X = Mass of N required for sludge production (mgN/d)

$$= \frac{f_{X,N} \cdot MX_v}{R_s} \tag{6.33f}$$

$f_{X,N}$ = Fraction of volatile solids which is nitrogen (0,1 mgN/mgVSS)
Y_{XA} = Autotroph specific yield (0,1 mgVSS/mgN)
b_A = Autotroph specific endogenous mass loss rate at temperature T° C
$= 0{,}04 \cdot (1{,}029)^{(T-20)}$ (/d) (6.33g)

- Ratio autotroph biological (active) mass/MLVSS

$$f_{X,BA} = MX_{BA}/MX_v \tag{6.34}$$

Calculation of $\hat{\mu}_A$*:* For both models

$$\frac{dN_{03}}{dt} = \frac{1}{Y_{XA}} \cdot \frac{dX_{BA}}{dt} = \frac{1}{Y_{XA}} \cdot \frac{\hat{\mu}_A \cdot N_a}{K_{SA} + N_a} \cdot X_{BA} \tag{6.35}$$

where

$\frac{dN_{03}}{dt}$ = Rate of change in nitrate concentration (mgN/ℓ/d)

$\frac{dX_{BA}}{dt}$ = Rate of change in autotroph concentration (mgVSS/ℓ/d)

N_a = Ammonia-N concentration (mgN/ℓ)

K_{SA} = Half saturation coefficient (mgN/ℓ)

In the batch test $N_a \gg K_{SA}$, $X_{BA} = X_{BA}BT$ and temperature = T°C, therefore

$$\frac{dN_{03}}{dt} \approx \frac{1}{Y_{XA}} \hat{\mu}_{AT} \cdot X_{BA}BT \tag{6.36}$$

and

$$\hat{\mu}_{AT} \approx \frac{dN_{03}}{dt} \cdot \frac{Y_{XA}}{f_{X,BA} \cdot X_v BT} \tag{6.37a}$$

where

$\hat{\mu}_{AT}$ = Maximum specific growth rate of autotrophs at T°C (/d)
$X_v BT$ = MLVSS concentration in the batch test (mgVSS/ℓ)
$= X_v \cdot V_{ml}/(V_{ml} + V_{ww})$ (6.37b)
X_v = MLVSS concentration of mixed liquor added to the batch test (mgVSS/ℓ)

Since in the batch test the nitrification rate virtually is constant, dN_{03}/dt can be approximated by $\Delta N_{03}/\Delta t$ which is the slope of the nitrate concentration time profile.

As input to the computer programs the value of $\hat{\mu}_A$ at 20°C is required. This is calculated from the value measured at T°C ($\hat{\mu}_{AT}$) as follows:

$$\hat{\mu}_{A20} = \hat{\mu}_{AT} \cdot \frac{1}{1{,}123^{(T-20)}} \tag{6.37c}$$

Calculation of $\hat{\mu}_H$: For both models

$$\frac{dX_{BH}}{dt} = Y_{XH} \cdot \frac{dS_{bs}}{dt} = \frac{Y_{XH}}{(1 - f_{cv} \cdot Y_{XH})} \cdot OUR_R \cdot 24 \qquad (6.38)$$

and

$$\frac{dX_{BH}}{dt} = \frac{\hat{\mu}_H \cdot S_{bs}}{K_{SH} + S_{bs}} \cdot X_{BH} \qquad (6.39)$$

where

$\frac{dX_{BH}}{dt}$ = Rate of synthesis of heterotroph biological (active) mass from RBCOD (mgVSS/ℓ/d)

$\frac{dS_{bs}}{dt}$ = Rate of change of RBCOD concentration (mgCOD/ℓ/d)

OUR_R = Oxygen utilization rate for RBCOD utilization (mgO/ℓ/h)
K_{SH} = Half saturation coefficient (mgCOD/ℓ)

Hence,

$$\frac{\hat{\mu}_H \cdot S_{bs}}{K_{SH} + S_{bs}} \cdot X_{BH} = \frac{Y_{XH}}{(1 - f_{cv} \cdot Y_{XH})} \cdot OUR_R \cdot 24$$

However, in the batch test during the initial peak OUR, $S_{bs} \gg K_{SH}$, $X_{BH} = X_{BH}BT$ and temperature = T°C, therefore

$$\hat{\mu}_{HT} = \frac{Y_{XH} \cdot OUR_R \cdot 24}{(1 - f_{cv} \cdot Y_{XH}) \cdot X_{BH}BT} = \frac{Y_{XH} \cdot OUR_R \cdot 24}{(1 - f_{cv} \cdot Y_{XH}) \cdot f_{X,BH} \cdot X_vBT} \qquad (6.40)$$

where

$\hat{\mu}_{HT}$ = Maximum specific growth rate of heterotrophs at T°C (/d)

In Eq (6.40), the OUR_R term is the OUR associated with utilization of RBCOD. In terms of the UCT model, this is obtained from the magnitude of the drop in the measured OUR when the RBCOD is depleted (see earlier and Fig 6.5). In terms of the IAWPRC Task Group model, this is obtained from the initial OUR minus the OUR for nitrification, (see Fig 6.5), i.e. for the IAWPRC model

$$OUR_R = \text{Peak OUR} - OUR_n \qquad (\text{mgO}/\ell/\text{h}) \qquad (6.41a)$$

where

OUR_n = OUR for nitrification

$$= 4{,}57 \frac{dN_{O3}}{dt} \qquad (\text{mgO}/\ell/\text{h}) \qquad (6.41b)$$

As noted earlier, dN_{03}/dt can be approximated by $\Delta N_{03}/\Delta t$ which is the slope of the nitrate concentration time profile.

As input to the computer programs the value of $\hat{\mu}_H$ at 20°C is required. This is calculated from the value measured at T°C ($\hat{\mu}_{HT}$) as follows:

$$\hat{\mu}_{H20} = \hat{\mu}_{HT} \cdot \frac{1}{1{,}20^{(T-20)}} \tag{6.42}$$

Calculation of RBCOD: The RBCOD concentration *in the wastewater added* to the batch test (S_{bsi}) can be estimated from the area under the peak OUR (Fig 6.5, Area 1).

$$S_{bsi} = \{ 1/(1-f_{cv} \cdot Y_{XH}) \} \cdot MO \cdot (V_{ml}+V_{ww})/V_{ww} \quad (mgCOD/\ell) \tag{6.43a}$$

where

MO = Mass of oxygen utilized in RBCOD consumption (i.e. Fig 6.5, Area 1)
$= \Delta OUR \cdot t$ (mgO/ℓ) (6.43b)
ΔOUR = Magnitude of drop in OUR from 1st to 2nd plateau $(mgO/\ell/h)$
t = Time 1st plateau lasts (h)

Also,

$$f_{ts} = S_{bsi}/S_{ti} \tag{6.44}$$

where

f_{ts} = Fraction of substrate total COD which is RBCOD
S_{ti} = Total COD concentration *of wastewater added* to the batch test $(mgCOD/\ell)$

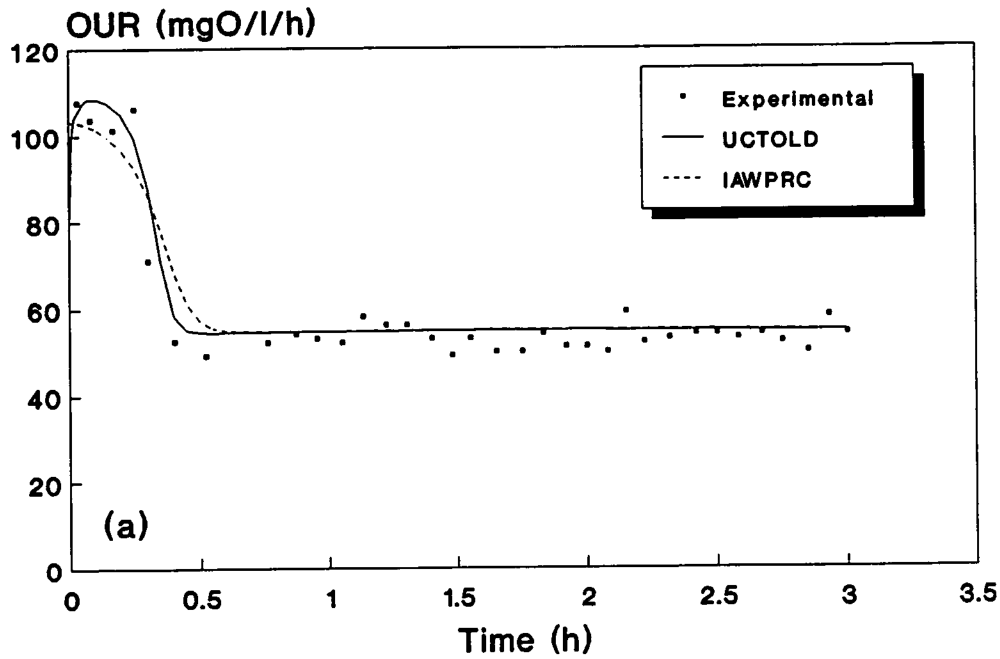
and

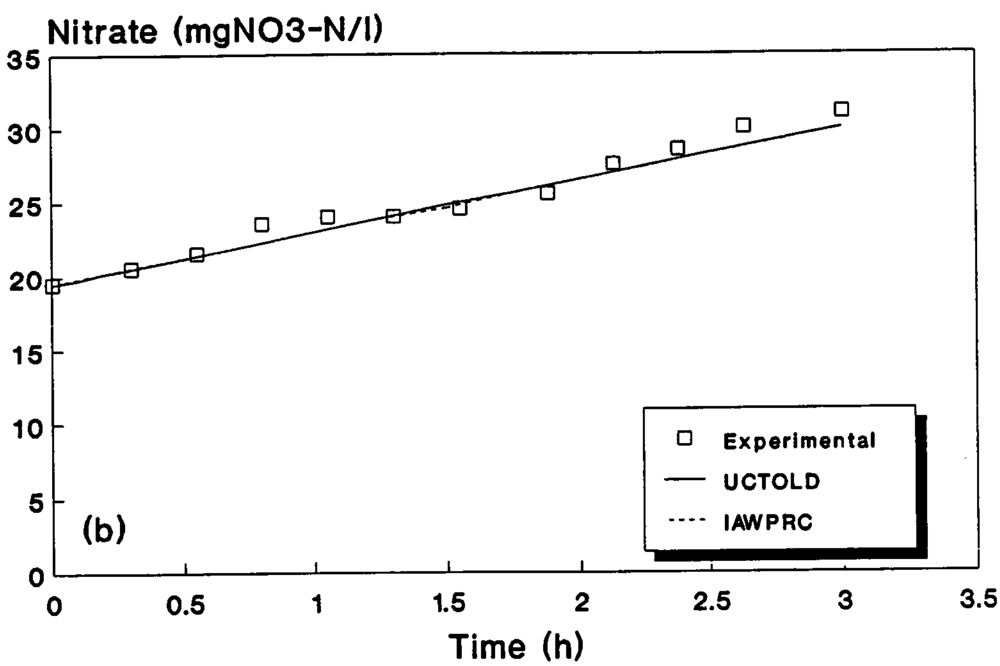
$$f_{bs} = S_{bsi}/S_{ti} \, (1 - f_{S,us} - f_{S,up}) \tag{6.45}$$

where

f_{bs} = fraction of biodegradable COD that is RBCOD.

Example calculation
Mixed liquor for the batch test was obtained from an intermittently fed fill-and-draw system operated at 20°C and with sludge age of 20 days. The volume of the reactor was 10ℓ and every day the system received 7500 mgCOD of Mitchell's Plain, South Africa, raw wastewater, which gave an influent TKN of 650 mgN/d. For the batch test, 10ℓ of mixed liquor with MLVSS = 2725 mgVSS/ℓ were drawn from the system and settled to 3,75ℓ with MLVSS = 2725·10/3,75 = 7267 mgVSS/ℓ. The mixed liquor was mixed with 6,25ℓ of Mitchell's Plain raw wastewater of 1159 mgCOD/ℓ and an aerobic batch test conducted at 20°C. The OUR, nitrate and TKN concentration time profiles are shown overleaf in Fig 6.8(a,b and c).





<u>**Fig 6.8**</u>: Predicted and experimental (a) OUR, (b) nitate, and (c) TKN time profiles in an aerobic batch test; predictions using batch versions of the **UCTOLD** and **IAWPRC** programs. Continued overleaf......

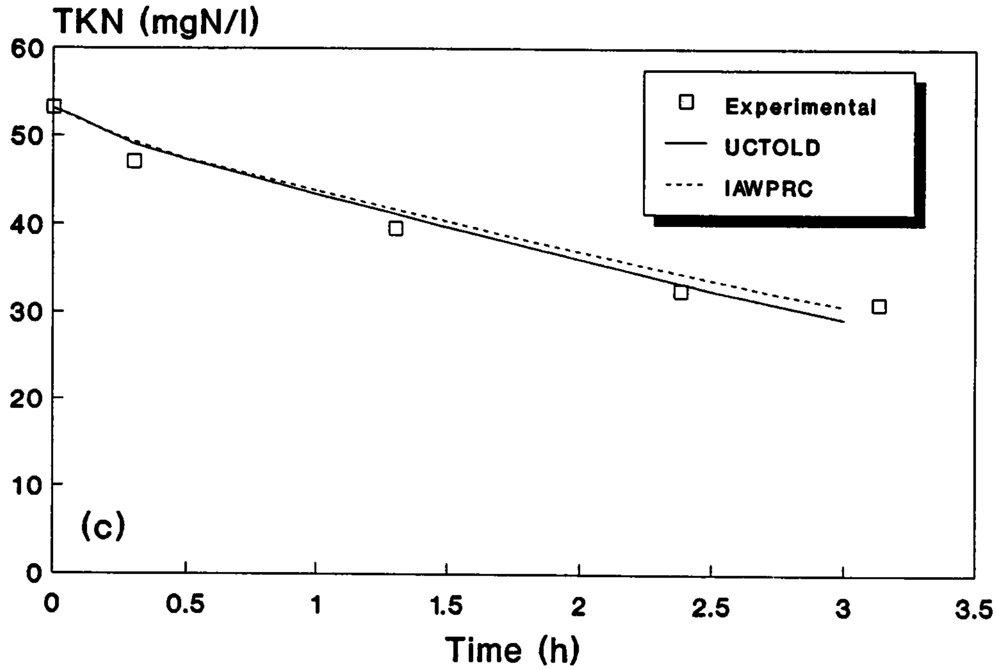


Fig 6.8: Predicted and experimental (a) OUR, (b) nitate, and (c) TKN time profiles in an aerobic batch test; predictions using batch versions of the **UCTOLD** and **IAWPRC** programs.

For this batch test,

$V_{ml} = 3{,}75\ell$
$V_{ww} = 6{,}25\ell$
$X_vBT = 7267 \cdot 3{,}75/(3{,}75+6{,}25) = 2725$ mgVSS/ℓ
$S_{ti} = 1159$ mgCOD/ℓ

The wastewater had been characterized previously using the procedures set out earlier, to give $f_{S,us} = 0{,}08$ and $f_{S,up} = 0{,}08$.

Parent system MLVSS fractions:

- From Eq (6.28)

$$MX_{BH} = \frac{7500\ (1 - 0{,}08 - 0{,}08) \cdot 0{,}45 \cdot 20}{(1 + 0{,}24 \cdot 20)}$$

$$= 9776 \text{ mgAVSS}$$

$$X_{BH} = 9776/10 = 978 \text{ mgAVSS}/\ell.$$

- From Eq (6.29)

$$MX_E = 0{,}2 \cdot 0{,}24 \cdot 20 \cdot 9776$$
$$= 9385 \text{ mg EVSS}$$

$$X_E = 9385/10 = 939 \text{ mgEVSS}/\ell$$

- From Eq (6.30)

$$MX_I = 0{,}08 \cdot 7500 \cdot 20/1{,}48$$
$$= 8108 \text{ mgIVSS}$$

$$X_I = 8108/10 = 811 \text{ mgIVSS}/\ell$$

- From Eq (6.31)

$$MX_v = 9776 + 9385 + 8108$$
$$= 27\,269 \text{ mgVSS}$$

$$X_v = 27\,269/10 = 2727 \text{ mgVSS}/\ell \text{ (measured to be 2725 mgVSS}/\ell)$$

- From Eq (6.32)

$$f_{X,BH} = 9776/27269 = 0{,}359$$

- From Eq (6.33c) and from the measured effluent TKN mass (MN_{te}) of 30 mgN/d

$$MN_{03} = 650 - 30 - 0{,}1 \cdot \frac{27269}{20}$$
$$= 484 \text{ mgN/d}$$

- From Eq (6.33a)

$$MX_{BA} = \frac{484 \cdot 0{,}1 \cdot 20}{1+0{,}04 \cdot 20}$$
$$= 538 \text{ mgVSS}$$

$$X_{BA} = 538/10 = 53{,}8 \text{ mgVSS}/\ell$$

- From Eq (6.34)

$$f_{X,BA} = 538/27269 = 0{,}02$$

Calculation of $\hat{\mu}_A$:

- From Fig 6.8b, the slope of the nitrate concentration time profile $\Delta N_{03}/\Delta t$ is (32-20,5)/3,13 = 3,674 mgN/ℓ/h = 88,2 mgN/ℓ/d. From Eq (6.37 a and b)

$$\hat{\mu}_{AT} = 88{,}2 \cdot \frac{0{,}1}{0{,}02 \cdot 2725} = 0{,}16/d$$

- From Eq (6.37c), adjusting $\hat{\mu}_{AT}$ to 20°C,

$$\hat{\mu}_{A20} = 0{,}16 \cdot \frac{1}{1{,}123^{(20-20)}}$$

$$= 0{,}16\ /d$$

Calculation of $\hat{\mu}_H$:

- In terms of the UCT model, from Fig 6.8a OUR_R = 116–52 = 64 mgO/ℓ/h. In terms of the IAWPRC Task Group model, from Fig 5.8a and Eq (6.41)

$$OUR_R = 116 - 4{,}57 \cdot 3{,}67$$

$$= 99\ mgO/\ell/h$$

- From Eq (6.40), for the UCT model

$$\hat{\mu}_{HT} = \frac{0.45 \cdot 64 \cdot 24}{(1-1{,}48 \cdot 0{,}45) \cdot 0{,}359 \cdot 2725}$$

$$= 2{,}12/d$$

- From Eq (6.42), adjusting $\hat{\mu}_{HT}$ to 20°C, for the UCT model,

$$\hat{\mu}_{H20} = 2{,}12 \cdot \frac{1}{1{,}20^{(20-20)}}$$

$$= 2{,}12\ /d$$

- From Eq (6.40), for the IAWPRC Task Group model,

$$\hat{\mu}_{HT} = \frac{0{,}45 \cdot 99 \cdot 24}{(1-1{,}48 \cdot 0{,}45) \cdot 0{,}359 \cdot 2725}$$

$$= 3{,}27/d$$

- From Eq (6.42), adjusting $\hat{\mu}_{HT}$ to 20°C, for the IAWPRC Task Group model,

$$\hat{\mu}_{H20} = 3{,}27 \cdot \frac{1}{1{,}20^{(20-20)}}$$

$$= 3{,}27\ /d$$

Calculation of RBCOD:

- From Eq (6.43b)

$$MO = 64 \cdot 0{,}25 = 16 \text{ mgO}/\ell$$

- From Eq (6.43a)

$$S_{bsi} = \frac{1}{(1-1{,}48 \cdot 0{,}45)} \cdot 16 \cdot (3{,}75+6{,}25)/6{,}25$$

$$= 76{,}7 \text{ mgCOD}/\ell$$

- From Eq (6.45)

$$f_{bs} = 76{,}7/1159(1 - 0{,}08 - 0{,}08)$$
$$= 0{,}08$$

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APPENDIX A

RETRIEVAL OF DIURNAL RESPONSE DATA

INTRODUCTION

One of the options in the diurnal sub-menu, **STORE DATA ON DISK,** allows the user to record the diurnal simulation results in a data file on disk. The purpose of this option is to enable off-line processing of the diurnal simulation results. For example, the user may wish to use a spreadsheet package such as *Lotus 1-2-3* or Borland's *Quattro* to plot results in a specific format. This Appendix explains the format of the data file and lists the parameters stored for the two programs **UCTOLD** and **IAWPRC.**

The program **RETRIEVE** on the 'Distribution disk' may be used to set up the data for off-line processing. Instructions on the use of the program also are provided in this Appendix.

The data are stored by the simulation programs in a Turbo Pascal *file of reals* with the extension .DID; the file name is specified by the user. The file consists of sets of values of the diurnal response parameters (concentrations), in each reactor, at each *Data interval* over the twenty four hour cycle. The number of sets of data depends on the *Data interval* used in the simulation (either 10, 15 or 30 minutes). For example, if the *Data interval* is set at 30 minutes in the simulation then $24 \cdot 60/30 + 1 = 49$ sets of values evenly spaced in time (every 30 minutes) from 0 to 24 hours will be stored. The default value for *Data Interval* is 15 minutes.

The number of diurnal response parameters (concentrations) differs for the two simulation programs; eighteen for **UCTOLD** and seventeen for **IAWPRC.** The parameters are listed separately below. For each program the size of the data file can be calculated as a check by the user, knowing the number of reactors (r), the *Data interval* (d), and that 6 bytes are used to store each real value. For example, if **UCTOLD** is used to simulate a two reactor system and the *Data interval* is set at 30 minutes, the size of the file will be

$$r \cdot 18 \cdot (24 \cdot 60/d + 1) \cdot 6 = 2 \cdot 18 \cdot 49 \cdot 6 = 10584 \text{ bytes}$$

This should correspond exactly to the file size obtained when the files on the disk are listed with the *dir* command in DOS.

PARAMETERS FOR UCTOLD

Concentrations of the following parameters are stored by the program UCTOLD

1.	Z_{BH}	- active heterotrophic biomass	(g COD m^{-3})
2.	Z_{BA}	- active autotrophic biomass	(g COD m^{-3})
3.	Z_E	- endogenous mass	(g COD m^{-3})
4.	Z_I	- inert mass	(g COD m^{-3})
5.	S_{ads}	- adsorbed slowly biodegradable substrate	(g COD m^{-3})
6.	S_{enm}	- enmeshed slowly biodegradable substrate	(g COD m^{-3})

7.	N_{obp}	- nitrogen organic biodegradable particulate	($g\ N\ m^{-3}$)
8.	S_{bs}	- soluble readily biodegradable substrate	($g\ COD\ m^{-3}$)
9.	N_a	- ammonia nitrogen	($g\ N\ m^{-3}$)
10.	N_{obs}	- nitrogen organic biodegradable soluble	($g\ N\ m^{-3}$)
11.	N_{o3}	- nitrate nitrogen	($g\ N\ m^{-3}$)
12.	Alk	- H_2CO_3* alkalinity	($moles\ m^{-3}$)
13.	S_{us}	- soluble unbiodegradable substrate	($g\ COD\ m^{-3}$)
14.	O_c	- carbonaceous oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
15.	O_n	- nitrification oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
16.	O_t	- total oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
17.	X_v	- volatile settleable solids (VSS)	($g\ VSS\ m^{-3}$)
18.	N_t	- total Kjeldahl nitrogen (TKN)	($g\ N\ m^{-3}$)

PARAMETERS FOR IAWPRC

Concentrations of the following parameters are stored by the program IAWPRC:

1.	Z_{BH}	- active heterotrophic biomass	($g\ COD\ m^{-3}$)
2.	Z_{BA}	- active autotrophic biomass	($g\ COD\ m^{-3}$)
3.	Z_E	- endogenous mass	($g\ COD\ m^{-3}$)
4.	Z_I	- inert mass	($g\ COD\ m^{-3}$)
5.	S_{enm}	- enmeshed slowly biodegradable substrate	($g\ COD\ m^{-3}$)
6.	N_{obp}	- nitrogen organic biodegradable particulate	($g\ N\ m^{-3}$)
7.	S_{bs}	- soluble readily biodegradable substrate	($g\ COD\ m^{-3}$)
8.	N_a	- ammonia nitrogen	($g\ N\ m^{-3}$)
9.	N_{obs}	- nitrogen organic biodegradable soluble	($g\ N\ m^{-3}$)
10.	N_{o3}	- nitrate nitrogen	($g\ N\ m^{-3}$)
11.	Alk	- H_2CO_3* alkalinity	($moles\ m^{-3}$)
12.	S_{us}	- soluble unbiodegradable substrate	($g\ COD\ m^{-3}$)
13.	O_c	- carbonaceous oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
14.	O_n	- nitrification oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
15.	O_t	- total oxygen utilization rate	($g\ O_2\ m^{-3}\ h^{-1}$)
16.	X_v	- volatile settleable solids (VSS)	($g\ VSS\ m^{-3}$)
17.	N_t	- total Kjeldahl nitrogen (TKN)	($g\ N\ m^{-3}$)

STRUCTURE OF THE DATA FILE

The Turbo Pascal code which writes the diurnal response results to the disk file is as follows:

```
for k := 1 to LastReactor do
  for j :=0 to DataPerDay do
    for i :=1 to NoDiVars do
      write (DATfile,Response [k]^ [i,j]) ;
```

where

DATfile	= Turbo Pascal *file of reals*
LastReactor	= number of reactors in system
DataPerDay	= index of last point at which data are stored in 24 hour cycle = 24 · 60 / *Data interval* (*Data interval* = 10, 15 or 30 minutes)
NoDiVars	= number of diurnal response parameters
Response	= three-dimensional array of concentration values

The data structure *Response* is an array of pointers to two dimensional matrices that store the concentration values for each reactor. The range of the two dimensional arrays is [1..*NoDiVars*,0..*DataPerDay*]. For example, if a *Data interval* of 30 minutes has been used in the simulation (and hence *DataPerDay* is 48) the matrix for a particular reactor may be visualized as follows:

$$\begin{bmatrix} (Z_{BH})_0 & (Z_{BA})_0 & \cdots\cdots\cdots & (X_V)_0 & (N_t)_0 \\ (Z_{BH})_1 & (Z_{BA})_1 & \cdots\cdots\cdots & (X_V)_1 & (N_t)_1 \\ \cdot & \cdot & \cdots\cdots\cdots & \cdot & \cdot \\ \cdot & \cdot & \cdots\cdots\cdots & \cdot & \cdot \\ \cdot & \cdot & \cdots\cdots\cdots & \cdot & \cdot \\ (Z_{BH})_{47} & (Z_{BA})_{47} & \cdots\cdots\cdots & (X_V)_{47} & (N_t)_{47} \\ (Z_{BH})_{48} & (Z_{BA})_{48} & \cdots\cdots\cdots & (X_V)_{48} & (N_t)_{48} \end{bmatrix}$$

The complete data file will consist of a set of such matrices, one for each reactor sequentially from the first to *LastReactor.* Each entry is a real number.

PROGRAM RETRIEVE FOR RETRIEVING DATA

The program **RETRIEVE** can be used to convert a data file with the extension **.DID** into a text file with the extension **.PRN** which is suitable for importing into a spread sheet package such as *Lotus 1-2-3.* The **.PRN** file is assigned the same name as the **.DID** file.

The program is executed by typing **RETRIEVE** at the DOS prompt. The program operates on files in the current directory of the logged drive. The user is then prompted to enter the name of the **.DID** file, the *Data interval* used in the simulation (10, 15 or 30 minutes), and the program used to create the data **(UCTOLD** or **IAWPRC)**. The program then converts the selected data file with extension **.DID** into a text file with extension **.PRN**.

The **.PRN** file is made up of a set of blocks of concentration values for each reactor. The blocks are stacked vertically and are separated by a blank line. Each block consists of (***DataPerDay*** + 1) lines. Each line consists of an integer denoting the reactor number, the corresponding time in the 24-hour cycle, followed by either 17 **(IAWPRC)** or 18 **(UCTOLD)** concentration values. The first line of the file provides headings for each column of values.

The **.PRN** file is loaded into the spreadsheets *Lotus 1-2-3* and Borland's *Quattro* by selecting following sequences of *commands:*

Lotus 1-2-3:	*Quattro:*
File	**File**
Import	**Import**
Numbers	**Comma & " " Delimited File**

In the case of *Lotus 1-2-3* the message **Part of file is missing** will appear at the lower left of the screen. However, the spreadsheet will appear intact when the **<Enter>** is pressed.

An example of how a part of a spreadsheet appears on the screen is shown below:

	A	B	C	D	E	F	G	H
1	REACTOR	TIME	Var 1	Var 2	Var 3	Var 4	Var 5	Var 6
2								
3	1	0	1067.343	29.873	265.214	566.893	46.858	3.574
4	1	0.25	1067.597	29.919	265.817	566.893	42.102	3.574
5	1	0.5	1067.379	29.962	266.417	566.893	38.068	3.584
6	1	0.75	1066.728	30.004	267.014	566.893	34.694	3.603
7	1	1	1065.693	30.045	267.609	566.893	31.903	3.636
8	1	1.25	1064.33	30.084	268.201	566.893	29.737	3.557
9	1	1.5	1062.682	30.122	268.792	566.893	27.988	3.497
10	1	1.75	1060.8	30.158	269.381	566.893	26.557	3.484
11	1	2	1058.729	30.193	269.967	566.893	25.402	3.492
12	1	2.25	1056.51	30.227	270.552	566.893	24.479	3.506
13	1	2.5	1054.174	30.259	271.136	566.893	23.738	3.525
14	1	2.75	1051.749	30.289	271.717	566.893	23.139	3.551
15	1	3	1049.254	30.318	272.297	566.893	22.648	3.585
16	1	3.25	1046.706	30.345	272.875	566.893	22.334	3.536
17	1	3.5	1044.114	30.37	273.452	566.893	22.089	3.49
18	1	3.75	1041.498	30.393	274.027	566.893	21.86	3.481
19	1	4	1038.861	30.415	274.6	566.893	21.65	3.495
20	1	4.25	1036.209	30.436	275.172	566.893	21.468	3.512

A1: ' REACTOR
14-Mar-91 02:00 PM READY

Using the spreadsheet package, the simulated data can be processed and plotted in a variety of ways. Also, any experimental data can be entered into the spreadsheet and plotted on the same graphs as the simulated data and more than one set of simulated data can be loaded into the spreadsheet and plotted on the same graph for comparison.

LISTING OF PROGRAM RETRIEVE

A listing of the program **RETRIEVE**, written in Turbo Pascal Version 4.0, can be found on the floppy disks labelled 'UCTOLD Listed Version' and 'IAWPRC Listed Version':

APPENDIX B

MODEL REPRESENTATION IN MATRIX FORMAT

INTRODUCTION

The two activated sludge system models included in the simulation package were presented in **Chapter 2**. In presenting the models, each was set out in matrix format. This Appendix provides a background and explanation of the use of the matrix format in the context of biological reactions (biological processes). A simple example is used to illustrate the technique.

MATHEMATICAL DESCRIPTION OF A MODEL

A comprehensive mathematical model for the simulation of biological system behaviour must account for a large number of reactions between a large number of components (*compounds*). In this manual, the reactions are referred to as *processes*, where processes act on certain *compounds* in the *system*, and convert these to other compounds. The set of distinct biological processes and the manner in which these act on the group of compounds constitute the biological or *process model*. The model should quantify, for each process, both the kinetics (rate-concentration dependence) and the stoichiometry (effect on the masses of compounds involved).

MATRIX METHOD FOR MODEL REPRESENTATION

An important part of biological modelling is that representation of the process model is clear and flexible. One convenient method of presentation is the matrix format.

The matrix method for model presentation described here is based on an approach used widely in chemical engineering kinetic modelling. In the context of biological systems, the method has been recommended by the IAWPRC Task Group on Mathematical Modelling in Wastewater Treatment. The matrix representation ensures clarity in presenting the processes in the model, their kinetics and their stoichiometric interactions with the compounds. In addition, it allows easy comparison of different models, and facilitates transforming the model into a computer program.

Setting up the matrix (process model)

Table B.1 presents, in matrix format, the essential components of a simple Monod-Herbert process model for aerobic microbial growth on a soluble substrate, accompanied by endogenous mass loss.

<u>Table B.1</u>: Monod-Herbert process model in matrix format.

COMPOUND i → j PROCESS ↓	1 Z_{BH}	2 S_{bs}	3 O	PROCESS RATE, ρ_j
1 GROWTH	1	$-1/Y_{ZH}$	$-(1-Y_{ZH})/Y_{ZH}$	$\hat{\mu}_H \frac{S_{bs}}{K_{SH}+S_{bs}} Z_{BH}$
2 DECAY	-1		-1	$b_H Z_{BH}$
OBSERVED CONVERSION RATES, $M\,L^{-3}\,T^{-1}$	$r_i = \sum \nu_{ij}\,\rho_j$			
	HETEROTROPH BIOMASS M (COD) L^{-3}	SOLUBLE SUBSTRATE M (COD) L^{-3}	OXYGEN M (-COD) L^{-3}	

The matrix is represented by a number of columns and rows; one column for each compound and one row for each process. The first step in setting up the matrix is to identify the compounds of relevance in the model. The Monod-Herbert model quantifies the growth of the heterotrophic biomass compound (Z_{BH}, COD units) at the expense of the soluble substrate compound (S_{bs}). By keeping track of Z_{BH} and S_{bs}, it is possible to calculate the oxygen requirement, in this fashion oxygen (O) can be included as a third compound (noting that oxygen has units of negative COD). The compounds are presented as symbols across the top of the table, and are defined (with dimensions) at the bottom of the corresponding matrix columns. The index i is assigned to the range of compounds. In this case, i ranges from 1 to 3 for the three compounds considered in this simple model.

The second step in developing the matrix is to identify the biological processes occurring in the system. These are conversions or transformations which affect the compounds considered in the model. Only two processes take place in this simple model - aerobic growth of organisms at the expense of soluble substrate, and endogenous mass loss. These are itemized one below the other at the left of the matrix. The index j is assigned to the range of processes. In this case, j can assume only a value of 1 or 2.

The process rates are formulated mathematically and are recorded down the right-hand side of the matrix in the appropriate row. These are given the symbol ρ_j with j denoting the index of the biological process.

Along each process row the stoichiometric coefficient for conversion from one compound to another is inserted so that each compound column lists the stoichiometric coefficients for the processes that influence that compound. The stoichiometric coefficients are given the symbol ν_{ij} where i denotes the index of the compound and j the index of the process. In our example for process 1, (aerobic growth of heterotrophs) the compound heterotrophic biomass (Z_{BH}) increases (+1), the compound soluble substrate (S_{bs}) decreases ($-1/Y_{ZH}$) and the compound oxygen (O) decreases [$-(1-Y_{ZH})/Y_{ZH}$].

The sign convention used in the matrix for each compound *per se* is 'negative for consumption' and 'positive for production'. Cognizance must be taken of the units used in the rate equations for the processes. For example, the rate equation for the process aerobic growth of heterotroph biomass, ρ_1, is written as a biomass growth rate (not as a substrate utilization rate) and has units of (mg cell COD growth)(mg substrate COD utilized)$^{-1}$d^{-1}. The stoichiometric values are thus normalized with respect to the heterotroph biomass concentration (Z_{BH}), i.e. the stoichiometric coefficients for Z_{BH} and S_{bs} are 1 and $-1/Y_{ZH}$ respectively, not Y_{ZH} and -1.

The stoichiometric coefficients, ν_{ij}, are greatly simplified by working in consistent units; in the example above, the compounds are expressed as COD equivalents. Provided consistent units are used, continuity may be checked from the stoichiometric parameters by moving across any row of the matrix. With consistent units, the sum of the stoichiometric coefficients must be zero (noting that oxygen is equivalent to negative COD). (In the matrices for **UCTOLD** and **IAWPRC** models, **Chapter 2**, the compound alkalinity is included. This compound does not constitute a part of the mass balance for the selected process and therefore must not be included in the continuity check.)

<u>System model</u>

The process model of the compounds, processes and rates presented in Table B.1 defines the behaviour at a single point in a system. To obtain the response of any system, the mass transport terms (e.g. mass flows and compounds into, and out of, each reactor) must be integrated with the process model, to give the *system model*. This is accomplished by setting up mass balance equations. An important benefit of the matrix representation is that it allows rapid and easy recognition of the fate of each compound, which aids in the preparation of the system mass balance equations.

The fundamental equation for a mass balance within any defined system boundary is:

$$\left\{\begin{matrix}\text{Rate}\\ \text{of}\\ \text{accumulation}\end{matrix}\right\} = \left\{\begin{matrix}\text{Rate}\\ \text{of}\\ \text{Input}\end{matrix}\right\} - \left\{\begin{matrix}\text{Rate}\\ \text{of}\\ \text{Output}\end{matrix}\right\} + \left\{\begin{matrix}\text{Rate of}\\ \text{Production}\\ \text{by Reaction}\end{matrix}\right\} \qquad \text{(B.1)}$$

The *input* and *output* terms are *transport terms* and depend upon the physical characteristics of the system being modelled. The *production by reaction* term (usually denoted by r_i for compound i) often is made up of a number of processes. In the matrix format, this information is obtained by summing the products of the stoichiometric coefficients, ν_{ij}, and the process rate expression, ρ_j, for the compound i being considered in the mass balance, i.e. moving down the matrix column for the specific compound i and accumulating the product of ν_{ij} and ρ_j :

$$r_i = \sum_j \nu_{ij}\, \rho_j \qquad \text{(B.2)}$$

For example, from Table B.1, the rate of reaction for the compound i = 1, heterotrophic biomass (Z_{BH}) at a point in the system would be:

$$r_1 = \frac{\hat{\mu}_H S_{bs}}{(K_{SH} + S_{bs})} Z_{BH} - b_H Z_{BH} \tag{B.3}$$

Similarly for the compound i = 2, soluble substrate (S_{bs}):

$$r_2 = -\frac{1}{Y_{ZH}} \frac{\hat{\mu}_H S_{bs}}{(K_{SH} + S_{bs})} Z_{BH} \tag{B.4}$$

and for compound i = 3, dissolved oxygen (O) :

$$r_3 = -\frac{(1-Y_{ZH})}{Y_{ZH}} \frac{\hat{\mu}_H S_{bs}}{(K_{SH} + S_{bs})} Z_{BH} - b_H Z_{BH} \tag{B.5}$$

To create the mass balance for any compound within a given system boundary (e.g. a completely mixed reactor), the conversion rate, r_i, would be multiplied by the reactor volume and added to the appropriate advective terms, i.e. input and output masses (where each mass is given by concentration times flow in or out of the system) for the particular system; this mass balance is not shown here as the system is not defined.

The rate of production by reaction, r_i, may be of interest on its own. For example, Eq (B.5) defines the 'rate of production' of O; therefore $-r_3$ defines the oxygen utilization rate at a point within the system. This parameter often is of interest in aerobic systems.

<u>Switching functions</u>

At this point, it is worth introducing an aspect of the kinetic expressions which often is useful and which is incorporated in the models presented in Chapter 2, namely *switching functions*. Consider the aerobic growth of biomass; in Table B.1 the Monod growth rate equation has been used:

$$\rho_1 = \frac{\hat{\mu}_H S_{bs}}{(K_{SH} + S_{bs})} Z_{BH} \tag{B.6}$$

In an environment where the dissolved oxygen concentration (O) is zero (or perhaps close to zero), the rate of this aerobic process also should be zero. Mathematically, this can be achieved by multiplying the Monod rate expression by a 'switching' factor which is zero when O is zero, and unity when the environment is 'fully' aerobic. Experience has shown that a switching function formulation that is very flexible and useful is one that takes the same form as the Monod formulation. For example, the switching function for oxygen is:

$$\frac{O}{(K_O + O)} \tag{B.7}$$

where K_O = switching constant of small magnitude (say 0,1 mgO ℓ^{-1})

The selection of a small value for K_O means that the value of the switching function decreases from near-unity to zero at very low O values, i.e. when the DO value decreases below about 0,2 mgO ℓ^{-1}. However, the function is mathematically continuous, which helps to eliminate problems of numerical instability in simulating system behaviour; such problems can arise if the rate is switched 'on' and 'off' discontinuously.

On incorporating the switching function the process rate equation then becomes:

$$\rho_1 = \frac{\hat{\mu}_H S_{bs}}{(K_{SH} + S_{bs})} \frac{O}{(K_O + O)} Z_{BH} \tag{B.8}$$

With the switching function operating on the Monod growth rate equation, when O is zero the value of the switching function is zero, and the process rate, ρ_1, will be zero. However, if O is say 1 mgO ℓ^{-1} then the value of the switching function is close to unity (i.e. 'fully' aerobic) and the process rate will be given by the Monod growth rate equation. In this way, the process of aerobic growth is switched 'on' or 'off' automatically by the model depending on the dissolved oxygen concentration.

In certain situations, the switching 'off' of one process may be linked to the switching 'on' of another. If, for example, the oxygen input to a nitrifying activated sludge system were terminated periodically, there would be a switch from aerobic to anoxic growth. The latter process is governed by kinetic and stoichiometric expressions which differ from those for the aerobic growth process. To account for this phenomenon in a single model, the rate equations for aerobic and anoxic growth can be multiplied by the appropriate switching functions as follows:

$$\rho_{aerobic} = \rho_{aerobic} \frac{O}{(K_O + O)} \tag{B.9}$$

$$\begin{aligned} \rho_{anoxic} &= \rho_{anoxic} \left\{ 1 - \frac{O}{(K_O + O)} \right\} \\ &= \rho_{anoxic} \frac{K_O}{(K_O + O)} \end{aligned} \tag{B.10}$$

In this instance, it is apparent that the selection of K_O will influence the point at which there is a switch from aerobic to anoxic growth, and *vice versa*. That is, K_O now influences the model predictions and is not only serving a mathematical objective. Therefore, whenever switching functions are used, care should be taken in the selection of the magnitude of the switching constant (K_O here) to ensure that the model predictions are not incorrectly biased.

APPENDIX C

REACTOR NUMBERING CONVENTION

The structure of the computer programs requires that the reactors in the system configuration are numbered using a defined procedure. This procedure should be followed when entering the **PLANT CONFIGURATION** data.

The reactor numbering procedure is as follows:

- *Draw the number of reactors in the system in a linear sequence following the stream of flow from reactor to reactor.*

- *The secondary clarifier is positioned at the downstream end of the reactor sequence.*

- *Number the reactors in descending order backwards, assigning the reactor immediately upstream of the clarifier with the highest number in the sequence. The reactor furthest upstream of the clarifier will then be assigned as number 1. The program assumes that inter-reactor flow passes sequentially from reactor to reactor (starting at No. 1) towards the clarifier without bypassing any reactor in the sequence.*

- *The mixed liquor recycles can have any reactor as point of origin but must discharge into a reactor upstream of the point of origin; that is, the reactor number of the point of origin must be greater than the reactor number of the point of discharge.*

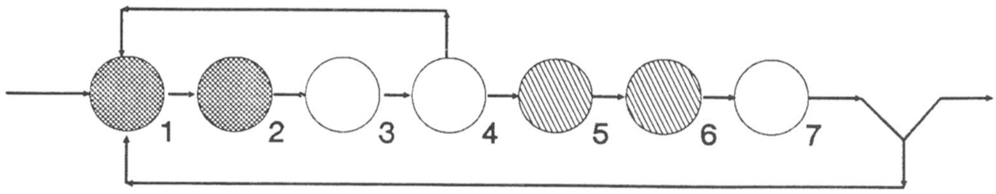
- *The influent flow can discharge into any reactor in the series, or be split between a number of reactors.*

- *The program allows simulation of systems with reactors in the underflow (RAS) recycle provided these reactors are at the start of the reactor sequence. The program cannot simulate systems with reactors between the point of origin and discharge of the mixed liquor recycles, i.e. no reactors are allowed in the mixed liquor recycles.*

Some examples illustrating the reactor numbering convention are set out below.

1. 4-STAGE BARDENPHO SYSTEM

This nitrification/denitrification (ND) system has 4 zones. Assuming that the system has been designed to comprise 2 reactors for each zone with the exception of the final reaeration zone, there is a total of 7 reactors in the series. Set up the 7 reactors in a linear sequence and number the reactors backwards from reactor 7 immediately upstream of the clarifier.



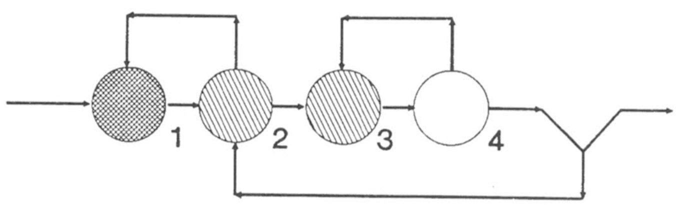
The sequence of reactors comprising the 4 zones is:

primary anoxic zone - reactors 1 and 2
main aeration zone - reactors 3 and 4
secondary anoxic zone - reactors 5 and 6
re-aeration zone - reactor 7.

In the 4-stage Bardenpho system the influent and clarifier underflow (RAS recycle) discharge to the start of the primary anoxic zone, i.e. reactor 1. The mixed liquor recycle has its point of origin at the end of the main aeration zone (reactor 4) and its point of discharge at the start of the primary anoxic zone (reactor 1).

2. MODIFIED UCT SYSTEM

This nitrification/denitrification/biological excess phosphorus removal (NDBEPR) system has 3 zones, anaerobic, primary anoxic, aerobic but the primary anoxic zone is divided in two. Therefore, for modelling purposes the system is comprised of 4 zones. Assuming there is one reactor for each zone, number these backwards from the clarifier as follows:



The sequence of reactors comprising the 4 zones is:

anaerobic zone - reactor 1
1st primary anoxic zone - reactor 2
2nd primary anoxic zone - reactor 3
aeration zone - reactor 4

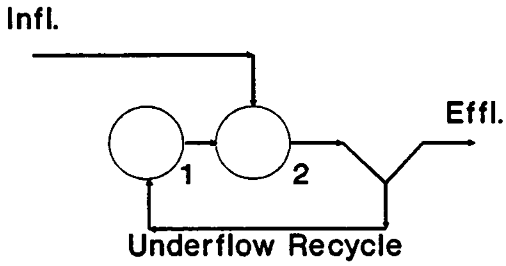
In the modified UCT system the underflow is discharged to the 1st primary anoxic zone (reactor 2) and the influent flow to the anaerobic zone (reactor 1). There are two mixed liquor recycles; one from the aerobic zone (reactor 4) to the 2nd primary anoxic zone (reactor 3) [a-recycle], and one from the 1st primary anoxic zone (reactor 2) to the anaerobic zone (reactor 1) [b- or r-recycle].

*Note: With NDBEPR systems, the **UCTOLD** and **IAWPRC** programs give only approximate estimates of the nitrification, denitrification and oxygen demand, and, do not give any estimates of phosphorus removal, see "**Scope of Models**".*

3. CONTACT STABILIZATION SYSTEM

In the first two examples the reactor numbering sequence is straightforward and does not differ from the way these systems usually are presented schematically. With the contact stabilization system the program reactor numbering sequence results in a configuration which may appear different from the usual schematic representation of the system.

The contact stabilization system has 2 zones, a contact reactor and a stabilization reactor. Assuming there is one reactor for each zone, number these backwards from the clarifier as follows:



1=Stabilization Reactor
2=Contact Reactor

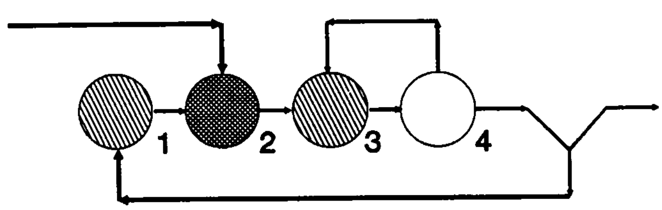
In the contact stabilization system there are no mixed liquor recycle flows – only an influent and an underflow stream (RAS recycle). The influent is discharged to the contact reactor which is directly upstream of the clarifier (reactor 2). The underflow discharges to the stabilization reactor (reactor 1). Hence the sequence of reactors comprising the 2 zones is:

stabilization reactor	– reactor 1
contact reactor	– reactor 2

From the example of the contact stabilization system, it is apparent that by discharging the influent into a reactor with a sequence number greater than 1, modelling of systems with reactors 'in' the underflow from the clarifier is possible. This approach is used for the Johannesburg system.

4. JOHANNESBURG (JHB) SYSTEM

This system is a nitrification/denitrification/biological excess phosphorus removal one with anaerobic, primary anoxic and aerobic zones and includes an additional anoxic zone in the underflow recycle. Assuming there is one reactor for each zone, number the 4 reactors in a linear sequence backwards from the clarifier as follows:



The sequence of reactors comprising the 4 zones is:

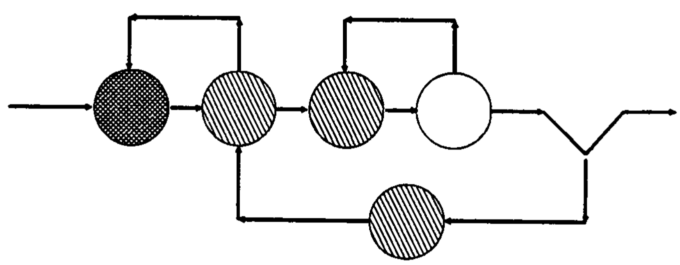
underflow anoxic zone	-	reactor 1
anaerobic zone	-	reactor 2
primary anoxic zone	-	reactor 3
aeration zone	-	reactor 4

In the Johannesburg system the underflow is discharged to the anoxic zone in the underflow stream (reactor 1). The discharge from this zone passes to the next reactor in the sequence (reactor 2); this is the anaerobic zone. The influent flow also discharges into the anaerobic zone (reactor 2). There is one mixed liquor recycle (a-recycle) in the system, from the aerobic zone (reactor 4) to the primary anoxic zone (reactor 3).

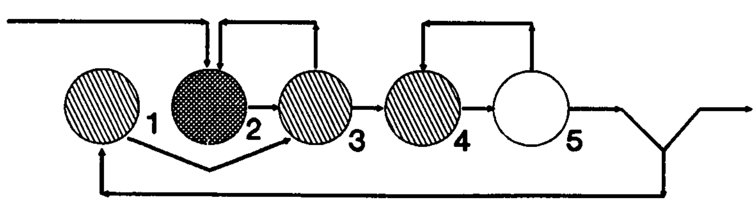
Note: The restrictions set out in the section on the modified UCT system above apply also to the Johannesburg system.

5. INADMISSIBLE CONFIGURATIONS

In general, if the outflow of a reactor/s in the underflow (usually reactor 1 or perhaps 1 and 2 in the sequence) discharges to a reactor with the next number in the sequence then the system can be simulated by the programs. This is demonstrated in the contact stabilization and Johannesburg system examples. However, if the outflow of the reactor/s in the underflow does not discharge to the reactor with the next sequence number, but to a reactor with a greater number, *the system cannot be simulated.* An example of such a case would be a 5-zone JHB/UCT combination system shown below:



Assuming one reactor per zone, drawing the 5 reactors in a linear sequence and numbering backwards from the clarifier yields:



The reactor immediately upstream of the clarifier is the aerobic zone (reactor 5). Continuing upstream from the aerobic zone, the next three reactors will be the 2nd and 1st primary anoxic reactors and the anaerobic reactor respectively, i.e. reactors 4, 3 and 2 respectively. Reactor 1 therefore must be the anoxic reactor in the underflow. However, discharge from this underflow anoxic reactor (reactor 1) must pass to the 1st primary anoxic zone (reactor 3); this would involve bypassing reactor 2 in the sequence. This is not possible with the structure of the programs - the outflow from reactor 1 can only be discharged to reactor 2. Consequently, the JHB/UCT type system, or any system where the outflow of a reactor does not pass directly to the next reactor in the sequence, cannot be simulated by the programs.

CAUTION

Simulations of the nitrification/denitrification/biological excess phosphorus removal (NDBEPR) system configurations (modified UCT and Johannesburg used as examples above) will be approximate only. These systems are configured to give biological excess phosphorus removal (BEPR) in addition to nitrification/denitrification (ND). However, the UCT and IAWPRC models do not incorporate the processes and compounds for BEPR. Also, the simulated nitrification, denitrification and oxygen demands are approximations only, although usually not greatly different from the observed behaviour.

APPENDIX D

SYMBOL SYSTEM

At the University of Cape Town (UCT) the first activated sludge model (a steady state model) was developed in 1969. Thereafter a series of models were developed, sequentially incorporating nitrification, denitrification, bisubstrate concept and death-regeneration (see **Chapter 2**). With each extension, the symbols previously defined had to be critically re-examined for consistency and for easy recognition.

With recent developments in the models, some new symbols had to be identified and, for consistency, some of the previous symbols had to be modified. Also, the symbol system had to be regularized to make provision for easy incorporation of future developments in the models, e.g. incorporation of biological excess phosphorus removal. The total list of revised symbols, and what they represent, is given in Table D.1.

Also given in Table D.1 are the equivalent symbols recommended by the IAWPRC task group. The reader will note that the symbols proposed by the UCT group are not the same as those recommended by the IAWPRC task group - the basis for naming the symbols differs sharply between the two symbol systems:

> In the UCT symbol system, subdivision of the compounds is on the basis of substrate (S), volatile mass (X–VSS units or Z–COD units), nitrogen (N), phosphorus (P) and oxygen (O). In the IAWPRC symbol system, subdivision of the compounds is on the basis of particulate (X) or soluble (S).

Irrespective of the merits of the two symbol systems, the UCT symbol system has been used because we believe it more readily allows recognition of the meaning of the symbols, and the symbols can be readily related to those used in publications by the UCT group over the past 20 years.

Table D.1: General UCT symbol system, updated to include biological excess phosphorus removal, with IAWPRC symbol equivalent.

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
	MAIN SYMBOLS	
S	Substrate	X or S
X	Volatile solids in VSS units	X
Z	Volatile solids in COD units	X
N	Nitrogen	X or S
P	Phosphorus	X or S
O	Oxygen	S_0
f	Fractional contents	f
M	Mass	–
	SUBSCRIPTS	
B	Biological (active) mass	B
E	Endogenous mass	E
I	Inert mass	I
H	Heterotrophs	H
A	Autotrophs (nitrifiers)	A
G	PolyP organisms	–
b	Biodegradable	–
u	Unbiodegradable	–
p	Particulate	–
s	Soluble	–
o	Organic (nitrogen)	–
a	Ammonia	–
i	Influent	–
e	Effluent	–
t	Total	–
	FRACTIONAL CONSTANTS	
$f_{X,YZ}$	Fraction of X which is YZ	i
$f_{XY,Z}$	Fraction of XY which is Z	

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
	COMPOUNDS	
S	**Substrate**	
S_u	Unbiodegradable	-
S_{up}	Unbiodegradable, particulate	X_I
S_{us}	Unbiodegradable, soluble	S_I
S_b	Biodegradable	-
S_{bs}	Biodegradable, readily (soluble)	S_S
$S_{bs,a}$	Biodegradable, readily (soluble), acetate	-
$S_{bs,c}$	Biodegradable, readily (soluble), complex	-
S_{bp}	Biodegradable, slowly (particulate)	X_S
S_{ads}	Adsorbed	-
S_{enm}	Enmeshed	X_S
S_{phb}	Stored PHB	-
Z	**Volatile solids (COD units)***	
Z_B	Biological (active) mass	-
Z_{BH}	Biological (active) mass, Heterotrophs	$X_{B,H}$
Z_{BA}	Biological (active) mass, Autotrophs	$X_{B,A}$
Z_{BG}	Biological (active) mass, PolyP organisms	-
Z_E	Endogenous mass	X_E
Z_I	Inert mass	X_I
Z_V	Total volatile solids (COD units) $= Z_B + Z_E + Z_I + (S_{ads} + S_{enm})$ $= X_v \cdot f_{cv}$	X_V

* Following the IAWPRC task group proposals volatile solids are expressed in COD units. The symbol X is substituted for Z if VSS units are used.

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
N	**Nitrogen**	
N_o	Organic	-
N_{ou}	Organic, unbiodegradable	-
N_{oup}	Organic, unbiodegradable, particulate	X_{NI}
N_{ous}	Organic, unbiodegradable, soluble	-
N_{ob}	Organic, biodegradable	-
N_{obp}	Organic, biodegradable, particulate	-
N_{obs}	Organic, biodegradable, soluble	S_{ND}
N_a	Ammonia	S_{NH}
N_{o3}	Nitrate	S_{NO}
N_Z	Nitrogen in volatile solids	-
N_{ZB}	Nitrogen in volatile solids, biological (active) mass	-
N_{ZE}	Nitrogen in volatile solids, endogenous mass	-
N_{ZI}	Nitrogen in volatile solids, inert mass	-
P	**Phosphorus**	
P_P	Particulate	X_P
P_S	Soluble	S_P
P_Z	Phosphorus in volatile solids	-
P_{ZB}	Phosphorus in volatile solids, biological (active) mass	-
P_{ZE}	Phosphorus in volatile solids, endogenous mass	-
P_{ZI}	Phosphorus in volatile solids, inert mass	-
P_{polyP}	Stored polyphosphate	-

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
$\underline{f}$	**FRACTIONAL CONTENTS**	f**
	Substrate	
$\underline{f_S}$,	Fraction of substrate, which is	
$f_{S,u}$	unbiodegradable	-
$f_{S,us}$	unbiodegradable, soluble	-
$f_{S,up}$	unbiodegradable, particulate	-
$f_{S,b}$	biodegradable	-
$f_{S,bs}$	biodegradable, readily (soluble)	-
$f_{S,bsa}$	biodegradable, readily (soluble), acetate	-
$f_{S,bsc}$	biodegradable, readily (soluble), complex	-
$f_{S,bp}$	biodegradable, slowly (particulate)	-
$f_{S,ZBH}$	biological active mass, heterotrophs	-
f_{bs}	Fraction of biodegradable substrate which is readily biodegradable	-
	Volatile solids (COD units)	
$\underline{f_Z}$,	Fraction of volatile solids, which is	
$f_{Z,B}$	biological (active) mass	-
$f_{Z,BH}$	biological (active) mass, heterotrophs	-
$f_{Z,BA}$	biological (active) mass, autotrophs	-
$f_{Z,BG}$	biological (active) mass, polyP organisms	-
$f_{Z,E}$	endogenous mass	-
$f_{Z,I}$	inert mass	-

** Little information is available in the literature on the subscripts for fractional contents, using the IAWPRC symbol system.

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
	Nitrogen	
$\underline{f_{N}}$,	Fraction of nitrogen, which is	
$f_{N,o}$	organic	–
$f_{N,ou}$	organic, unbiodegradable	–
$f_{N,ous}$	organic, unbiodegradable, soluble	–
$f_{N,oup}$	organic, unbiodegradable, particulate	–
$f_{N,ob}$	organic, biodegradable	–
$f_{N,obs}$	organic, biodegradable, soluble	–
$f_{N,obp}$	organic, biodegradable, particulate	–
$f_{N,a}$	ammonia	–
$f_{Nob,p}$	Fraction of organic biodegradable nitrogen which is particulate	–
	Phosphorus	
$\underline{f_{P}}$,	Fraction of phosphorus, which is	
$f_{P,p}$	particulate	–
$f_{P,s}$	soluble	–

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
	Nitrogen content	
$f_{,\underline{N}}$	Fraction of "......", which is nitrogen	
$f_{Z,N}$	volatile solids (COD units)	-
$f_{ZB,N}$	volatile solids (COD units), biological active mass	i_{XBN}
$f_{ZE,N}$	volatile solids (COD units), endogenous mass	i_{XEN}
$f_{ZI,N}$	volatile solids (COD units), inert mass	-
	Phosphorus content	
$f_{,\underline{P}}$	Fraction of "......", which is phosphorus	
$f_{Z,P}$	volatile solids (COD units)	-
$f_{ZB,P}$	volatile solids (COD units), biological (active) mass	i_{XBP}
$f_{ZE,P}$	volatile solids (COD units), endogenous mass	i_{XEP}
$f_{ZI,P}$	volatile solids (COD units), inert mass	-
	Endogenous residue	
$f_{\underline{E}}$	Fraction of the organism that remains as unbiodegradable residue	f_E
	Special fractional contents	
f_{cv}	COD/VSS ratio	
f_i	MLVSS/MLSS ratio	
f_{NS}	Influent TKN/COD concentration ratio	

Table D.1: (Continued)

UCT SYMBOL SYSTEM		IAWPRC
Symbol	Description	Equivalent
	CONSTANTS* **	
$\underline{Y}$	<u>True specific yield (COD units)</u>	Y
Y_{ZH}	Heterotrophs	Y_H
Y_{ZA}	Autotrophs	Y_A
$\underline{\hat{\mu}}$	<u>Maximum specific growth rate</u>	μ_{max}
$\hat{\mu}_H$	Heterotrophs	$\hat{\mu}_H$
$\hat{\mu}_A$	Autotrophs	$\hat{\mu}_A$
K_S	<u>Monod half saturation coefficient</u>	K_S
K_{SH}	Heterotrophs	K_{SH}
K_{SA}	Autotrophs	K_{SA}
R_h	Hydraulic retention time	Θ
R_s	Sludge age - solids retention time	Θ_c
Q	Flow rate	Q
V	Volume	V

*** Only constants common to both bacterial populations are given. A number of constants exist specific to each population. Constants used in this manual are given in the list of symbols, and in the relevant chapters.